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Chapter 1

Keeping it Real

1.1 Exercise 1.1

Explain the error in the following “proof” that $2 = 1$.

Let $x = y$. Then

$$\begin{aligned}x^2 &= xy \\x^2 + y^2 &= xy - y^2 \\(x + y)(x - y) &= y(x - y) \\x + y &= y \\2y &= y \\2 &= 1\end{aligned}$$

Solution

Between $(x + y)(x - y) = y(x - y)$ and $x + y = y$, we divide by $x - y$, which from the given, is equal to zero. Hence, the step involves dividing by zero, itself undefined and not allowed to be performed.

1.2 Exercise 1.2

Which of the following statements are true? Give a short explanation for each of your answers?

- (a) For every $n \in \mathbb{N}$ there is an $m \in \mathbb{N}$ such that $m > n$.
- (b) For every $m \in \mathbb{N}$ there is an $n \in \mathbb{N}$ such that $m > n$.
- (c) There is an $m \in \mathbb{N}$ such that for every $n \in \mathbb{N}$, $m \geq n$.
- (d) There is an $n \in \mathbb{N}$ such that for every $m \in \mathbb{N}$, $m \geq n$.
- (e) There is an $n \in \mathbb{R}$ such that for every $m \in \mathbb{R}$, $m \geq n$.
- (f) For every pair $x < y$ of integers, there is an integer z such that $x < z < y$.
- (g) For every pair $x < y$ of real numbers, there is a real number z such that $x < z < y$.

1.2.1 Solution (a) True

Proof. We begin with our given, which is that $n \in \mathbb{N}$. Natural numbers have a quality that they are ordered such that successive numbers are a distance of 1 apart. Natural numbers are also unlimited in the direction of positive infinity. This means that for every $n \in \mathbb{N}$, there is a greater natural number. In other words, for every $n \in \mathbb{N}$, there exists an $m \in \mathbb{N}$ such that $m > n$. OE Δ

1.2.2 Solution (b) False

To prove this false, we need but find a counterexample. For the counterexample, we look at the number 1. $1 \in \mathbb{N}$. However, there is no $n \in \mathbb{N}$ such that $1 > n$. As such, it is false.

1.2.3 Solution (c) False

There is not maximum number in the natural numbers. For this to be true, it would have to contradict part (a), which has been proven. As such it is false.

1.2.4 Solution (d) True

The contrapositive of this is there is no $n \in \mathbb{N}$ such that for every $m \in \mathbb{N}$, $m < n$. Since there are infinitely many $\mathbb{R}$ with no upward bound, there will

be no number $n \in \mathbb{N}$ such that $m < n$ for every $m \in \mathbb{N}$. This makes it true.

1.2.5 Solution (e) True

The contrapositive of this is there is no $n \in \mathbb{R}$ such that for every $m \in \mathbb{R}$, $m < n$. Since there are infinitely many $\mathbb{R}$ with no upward bound, there will be no number $n \in \mathbb{R}$ such that $m < n$ for every $m \in \mathbb{R}$. This makes it true.

1.2.6 Solution (f) False

If x and y are adjacent, like 0 and 1, then the number z such that $x < z < y$ would have to be of distance less than 1 from the two adjacent numbers. This contradicts the principle of integers, which says that all integers are of distance 1 from their adjacent numbers. The statement is false.

1.2.7 Solution (g) True

There are considered to be infinitely many real numbers between two other real numbers. This means that for any $x, y \in \mathbb{R}$, there does exist $z \in \mathbb{R}$ such that $x < z < y$.

1.3 Exercise 1.3

If A and B are two boxes (possibly with things inside), describe the following in terms of boxes:

$$(a) A \setminus B$$

$$(b) \mathcal{P}(A)$$

$$(c) |A|$$

1.3.1 Solution (a)

$A \setminus B$ means the set of objects in A not in B . $A \setminus B = \{x : x \in A \text{ and } x \notin B\}$. In terms of boxes, this would mean the box with everything that is in A but not in B .

1.3.2 Solution (b)

$\mathcal{P}(A)$ or the *power set* of A means the set of all subsets of A . $\mathcal{P}(A) = \{X : X \subseteq A\}$. In terms of boxes, this would be the box containing every box containing a combination of only different objects contained in A .

1.3.3 Solution (c)

$|A|$ or the cardinality of A means the count of all elements in A . In terms of boxes, this would be the number of objects contained in the box A .

1.4 Exercise 1.4

If $A_1, A_2, A_3, \dots, A_n$ are all boxes (possibly with things inside), describe the following in terms of boxes:

$$(a) \bigcup_{i=1}^n A_i = A_1 \cup A_2 \cup \dots \cup A_n \qquad (b) \bigcap_{i=1}^n A_i = A_1 \cap A_2 \cap \dots \cap A_n$$

1.4.1 Solution (a)

A box containing all the objects in each of the boxes $A_1, A_2, A_3, \dots, A_n$.

1.4.2 Solution (b)

A box containing each object that is in every one of the boxes $A_1, A_2, A_3, \dots, A_n$.

1.5 Exercise 1.5

Prove that each of the following holds for any sets A and B .

- (a) $A \cup B = A$ if and only if $B \subseteq A$.
- (b) $A \cap B = A$ if and only if $A \subseteq B$.
- (c) $A \setminus B = A$ if and only if $A \cap B = \emptyset$.
- (d) $A \setminus B = \emptyset$ if and only if $A \subseteq B$.

1.5.1 Solution (a)

We have to prove $A \cup B = A$ if and only if $B \subseteq A$. To prove this, we must prove that $A \cup B = A \implies B \subseteq A$ and $B \subseteq A \implies A \cup B = A$. We begin with the first.

Proof. We know that $A \cup B = A$. This means that for any $b \in B$ or $b \in A$, $b \in A$. Taking the first half of this, we get $b \in B \implies b \in A$. This can be rewritten as $B \subseteq A$. QED OEΔ

Now we prove the second.

Proof. We know that $B \subseteq A$. This can be written that $b \in A$ for every $b \in B$, or $\frac{b \in B}{b \in A}$.

From identity, we also know that $b \in A$ for every $b \in A$, or $\frac{b \in A}{b \in A}$.

Putting these together, we get $b \in A$ for every $b \in A$ or $b \in B$, rewritten as $\frac{(b \in A) \vee (b \in B)}{b \in A}$.

Rewriting this in sets, we get $\frac{\{b: b \in A \text{ or } b \in B\}}{\{b: b \in A\}}$.

Putting this in implies and sets notation, we finally end up with $A \cup B = A$. QED OEΔ

Putting these together, we get the if and only if statement we were looking for.

1.5.2 Solution (b)

Part 1: $A \subseteq B \implies A \cap B = A$

Proof. (1) It is given that $A \subseteq B$. From this, we know from definitional that for any $a \in A$, $a \in B$.

$$a \in A \implies a \in B$$

(2) In what I call “The 30s conjecture” (iff a person is in their thirties then they are in their thirties and are in their thirties), we can say that if and only if $a \in A$ and $a \in A$, then $a \in A$.

$$A = \{a : a \in A, a \in A\}$$

(3) From step (1), we can put in the implication into part (2) that if and only if $a \in A$ and $a \in B$, then $a \in A$.

$$A = \{a : a \in A, a \in B\}$$

(4) We can rewrite that as $A \cap B = A$. QED OEΔ

Part 2: $A \cap B = A \implies A \subseteq B$

Proof. Suppose that $A \cap B = A$. We can establish definitions for this.

$$A \cap B = \{x : x \in A \text{ and } x \in B\} \tag{1.1}$$

From the transitive property, we can continue.

$$A = \{x : x \in A \text{ and } x \in B\} \tag{1.2}$$

We can then establish the reflexive definition for the set A and apply that to the above equation with the transitive property.

$$A = \{x : x \in A\} \tag{1.3}$$

$$\{x : x \in A\} = \{x : x \in A \text{ and } x \in B\} \tag{1.4}$$

We can use this to define a generic element x .

$$x \in A \Leftrightarrow x \in A \text{ and } x \in B \tag{1.5}$$

$$x \in A \implies x \in A \text{ and } x \in B \tag{1.6}$$

This can be separated into two implications.

$$x \in A \implies x \in A \tag{1.7}$$

$$x \in A \implies x \in B \tag{1.8}$$

We can turn the second point into $A \subseteq B$.
Q.E.D. OEΔ

1.5.3 Solution (c)

Part 1: $A \cap B = \emptyset \implies A \setminus B = A$

Proof. If $A \cap B = \emptyset$, that means that $a \notin B$ for any $a \in A$ and $a \notin A$ for any $a \in B$. It also means that A and B are disjoint.

$$a \in B \implies a \notin A \quad (1.9)$$

$$a \in A \implies a \notin B \quad (1.10)$$

We can also define the set $A \setminus B$.

$$A \setminus B = \{x : x \in A, x \notin B\} \quad (1.11)$$

We can set the falsehood to its equivalent.

$$A \setminus B = \{x : x \in A, \overline{x \in B}\} \quad (1.12)$$

The equivalent takes in (1.2).

$$A \setminus B = \{x : x \in A, \overline{x \notin A}\} \quad (1.13)$$

We can turn the double negative into a positive.

$$A \setminus B = \{x : x \in A, x \in A\} \quad (1.14)$$

We can then combine the two and compare then to the identity.

$$A \setminus B = \{x : x \in A\} \quad (1.15)$$

$$A = \{x : x \in A\} \quad (1.16)$$

$$A \setminus B = A \quad (1.17)$$

QED

OEΔ

Part 2: $A \setminus B = A \implies A \cap B = \emptyset$

Proof. Given: $A \setminus B = A$.

$$x \in A \wedge x \notin B \leftrightarrow x \in A \quad (1.18)$$

We define the set $A \cap B$.

$$A \cap B = \{x : x \in A, x \in B\} \quad (1.19)$$

We can substitute in the equivalence (1.10) into (1.11).

$$A \cap B = \{x : x \in A \wedge x \notin B, x \in B\} \quad (1.20)$$

$$A \cap B = \{x : x \in A, x \notin B, x \in B\} \quad (1.21)$$

This then gives us a contradiction. An element cannot both be in and not in set B . As such, we can equate the equivalence with an empty set.

$$A \cap B = \emptyset \quad (1.22)$$

QED

OEΔ

1.5.4 Solution (d)

Part 1: $A \subseteq B \implies A \setminus B = \emptyset$.

Proof. We can define the relationship $A \subseteq B$ and the set $A \setminus B$.

$$x \in A \implies x \in B \quad (1.23)$$

$$A \setminus B = \{x : x \in A, x \notin B\} \quad (1.24)$$

We can substitute (1.15) into (1.16).

$$A \setminus B = \{x : x \in B, x \notin B\} \quad (1.25)$$

This generates a contradiction, which we can use to turn the set into a null set.

$$A \setminus B = \emptyset \quad (1.26)$$

QED

OEΔ

Part 2: $A \setminus B = \emptyset \implies A \subseteq B$.

Proof by Contradiction. For the sake of contradiction, we assume that $A \not\subseteq B$. We also have as given, that $A \setminus B = \emptyset$.

$$A \setminus B = \emptyset \quad (1.27)$$

$$A \setminus B = \{x : x \in A, x \notin B\} \quad (1.28)$$

We can infer from this the existence of an element x .

$$x \in A \quad (1.29)$$

$$x \notin B \quad (1.30)$$

Based on the definition, it would follow that x would be an element of $A \setminus B$. We can apply a transitive property to this.

$$x \in A \setminus B \tag{1.31}$$

$$x \in \emptyset \tag{1.32}$$

Since $\emptyset$ has no elements, it contradicts its definition to say that x is an element of $\emptyset$. As such, our assumption cannot be true. This validates that $A \setminus B = \emptyset \implies A \subseteq B$.

Q.E.D.

OE Δ

1.6 Exercise 1.6

Suppose $f: X \rightarrow Y$ and $A \subseteq X$ and $B \subseteq Y$.

- (a) Prove that $f(f^{-1}(B)) \subseteq B$.
- (b) Give an example where $f(f^{-1}(B)) \neq B$.
- (c) Prove that $A \subseteq f^{-1}(f(A))$.
- (d) Give an example where $A \neq f^{-1}(f(A))$.

1.6.1 Solution (a)

Proof. We can set a reflexive truth involving B .

$$B = B \tag{1.33}$$

Since $f: X \rightarrow Y$ and $B \subseteq Y$, we can say that B is part of the codomain of f . We can as such apply the inverse function of f to both sides of the reflexive property.

$$f^{-1}(B) = f^{-1}(B) \tag{1.34}$$

Lastly, we can apply the function f to both sides, which may cancel out on the right side. Since there may be some values in $b \in B$ for which $f^{-1}(B)$ is undefined or does not exist, we can no longer set it as an equality. However, we can set it as a subset equality.

$$f(f^{-1}(B)) \subseteq B \tag{1.35}$$

QED

OEΔ

1.6.2 Solution (b)

$$f(x) = \frac{1}{x} \tag{1.36}$$

1.6.3 Solution (c)

Proof. We will first define the function f and the function $f^{-1} \circ f$.

$$f(A) = \{f(x)|x \in A\} \quad (1.37)$$

$$f^{-1}(f(A)) = \{f^{-1}(x)|x \in f(A)\} \quad (1.38)$$

From this, we can assume the existence of a generic element x that is an element of A .

$$x \in A \quad (1.39)$$

Since $x \in A$, we can then apply the function f to both sides.

$$f(x) \in f(A) \quad (1.40)$$

We can then apply the definition of the image to this.

$$x \in f^{-1}(f(A)) \quad (1.41)$$

Since x is a generic element of A , we have shown that every element of A is in $f^{-1}(f(A))$.

$$A \subseteq f^{-1}(f(A)) \quad (1.42)$$

QED

OEΔ

1.6.4 Section (d)

$$f(x) = \sqrt{x}$$

1.7 Exercise 1.7

Suppose that $f: X \rightarrow Y$ and $g: Y \rightarrow X$ are functions and that the composite $g \circ f$ is the identity function $\text{id}: X \rightarrow X$. (The identity function sends every element to itself: $\text{id}(x) = x$.) Show that f must be a one-to-one function and that g must be an onto function.

Solution

First, we have to prove that f must be a one-to-one function.

Proof. First, we know that $g \circ f(x) = g(f(x)) = x$. Suppose that we have two values $x, y \in X$ and that they have the same value when the function f is applied to them.

$$f(x) = f(y) \tag{1.43}$$

We can use the function g on both sides of this.

$$g(f(x)) = g(f(y)) \tag{1.44}$$

Since $g \circ f = \text{id}$, we replace this.

$$\text{id}(x) = \text{id}(y) \tag{1.45}$$

This is equivalent to our ending point.

$$x = y \tag{1.46}$$

Putting it all together, we get an implication.

$$f(x) = f(y) \implies x = y \tag{1.47}$$

Q.E.D.

OEΔ

Now, we have to prove that g is an *onto* function. As a reminder, a function $f: A \rightarrow B$ is onto (or surjective) if for every $b \in B$, there exists some $a \in A$ such that $f(a) = b$.

Proof by Contradiction. For contradiction, suppose that $\exists x \in X$ such that for $\forall y \in Y$, $g(y) \neq x$. The given information we have is that $g \circ f$ is the

identity function id . By its very nature, id is always bijective. From that, we can establish that it applies to our value x , as does its equivalent.

$$\exists x \in X : x = \text{id}(x) \quad (1.48)$$

$$\exists x \in X : x = g(f(x)) \quad (1.49)$$

We can apply this to the given inequality.

$$g(y) \neq x \quad (1.50)$$

$$g(y) \neq g(f(x)) \quad (1.51)$$

Since $f : X \rightarrow Y$ and $x \in X$, we can establish that $\exists y \in Y : y = f(x)$.

$$\exists y \in Y : f(x) = y \quad (1.52)$$

We can take the final contradiction assumption, creating new values appropriately and apply substitution to it. Since the value of y can be any value in Y , that value could be $f(x)$, which is established to exist.

$$\exists x \in X : g(y) \neq x \quad (1.53)$$

$$\exists x \in X : g(f(x)) \neq x \quad (1.54)$$

$$\exists x \in X : \text{id}(x) \neq x \quad (1.55)$$

This means that x is not equal to its identity function. This contradicts the sheer fact of $\text{id}(x) = x$. This shows that $\text{id}(x) = x$ can only be true if our initial assumption is false. This proves our point by contradiction.

Q.E.D.

OEΔ

1.8 Exercise 1.8

The following are special cases of De Morgan's laws.

- (a) Prove that $(A \cap B)^c = A^c \cup B^c$.
- (b) Prove that $(A \cup B)^c = A^c \cap B^c$.

1.8.1 Solution (a)

Prove that $(A \cap B)^c = A^c \cup B^c$.

Proof by Truth Table. We will have all the following conditions apply for whether a given element x is part of it.

A	B	$A \cap B$	$(A \cap B)^c$	A	B	A^c	B^c	$A^c \cup B^c$
T	T	T	F	T	T	F	F	F
T	F	F	T	T	F	F	T	T
F	T	F	T	F	T	T	F	T
F	F	F	T	F	F	T	T	T

Since the columns of $(A \cap B)^c$ and $A^c \cup B^c$ are the same for the same values of A and B in their respective columns, we can conclude that $(A \cap B)^c = A^c \cup B^c$.

Q.E.D.

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1.8.2 Solution (b)

Prove that $(A \cup B)^c = A^c \cap B^c$.

Proof by Truth Table. We will have all the following conditions apply for whether a given element x is part of it.

A	B	$A \cup B$	$(A \cup B)^c$	A	B	A^c	B^c	$A^c \cap B^c$
T	T	T	F	T	T	F	F	F
T	F	T	F	T	F	F	T	F
F	T	T	F	F	T	T	F	F
F	F	F	T	F	F	T	T	T

Since the columns of $(A \cup B)^c$ and $A^c \cap B^c$ are the same for the same values of A and B in their respective columns, we can conclude that $(A \cup B)^c = A^c \cap B^c$.

Q.E.D.

OEΔ

1.9 Exercise 1.9

- (a) Prove that $\sqrt{3}$ is irrational.
- (b) What goes wrong when you try to adapt your argument from part (a) to show that $\sqrt{4}$ is irrational (which is absurd)?
- (c) In part (a) you proved that $\sqrt{3}$ to be irrational, and essentially the same proof shows that $\sqrt{5}$ is irrational. By considering their product or otherwise, prove that $\sqrt{3} + \sqrt{5}$ and $\sqrt{3} - \sqrt{5}$ are either both rational or both irrational. Deduce that they must both be irrational.

1.9.1 Solution (a)

Proof that $\sqrt{3}$ is irrational.

Proof by Contradiction. Suppose $\sqrt{3}$ is rational. This would mean that for some relatively prime $p, q \in \mathbb{Z}$, $\sqrt{3} = \frac{p}{q}$. Square both sides and solve for p^2 .

$$\sqrt{3} = \frac{p}{q} \tag{1.56}$$

$$3 = \frac{p^2}{q^2} \tag{1.57}$$

$$p^2 = 3q^2 \tag{1.58}$$

By Euclid's Lemma, we know that if 3 divides p^2 , then 3 divides p . As such, we can substitute in $p = 3j$ for some $j \in \mathbb{Z}$. We can continue with prior steps.

$$(3j)^2 = 3q^2 \tag{1.59}$$

$$3q^2 = 9j^2 \tag{1.60}$$

$$q^2 = 3j^2 \tag{1.61}$$

This tells us that 3 divides q^2 and by Euclid's Lemma that 3 divides q . This means that 3 divides both p and q . This means that p and q are not relatively prime, which contradicts our assumption. Therefore, $\sqrt{3}$ has to be irrational.

Q.E.D.

OEΔ

1.9.2 Solution (b)

Proof.

$$\sqrt{4} = \frac{p}{q} \quad (1.62)$$

$$4 = \frac{p^2}{q^2} \quad (1.63)$$

$$p^2 = 4q^2 \quad (1.64)$$

Now, the issue we have here is that Euclid's lemma only works for prime numbers. Since 4 is not prime, we can only say that if 4 divides p^2 , then $\sqrt{4} = 2$ divides p . Here is what happens if we try to substitute in $p = 2j$ for some $j \in \mathbb{Z}$.

$$(2j)^2 = 4q^2 \quad (1.65)$$

$$4j^2 = 4q^2 \quad (1.66)$$

$$j^2 = q^2 \quad (1.67)$$

We can substitute this into our other equations. However, we would be going in circles and would be unable to arrive at a contradiction. This proves that we cannot prove that $\sqrt{4}$ is irrational the way we did for $\sqrt{3}$. $\square$

1.9.3 Solution (c)

First we prove they are both either rational or irrational.

Proof by Contradiction. We first consider the product of the two.

$$(\sqrt{3} + \sqrt{5})(\sqrt{3} - \sqrt{5}) = 3 - 5 = -2 \quad (1.68)$$

We can clearly see that the product of the two is a rational number. Now, suppose (for contradiction purposes) that one and only one of two multiplied numbers is rational. That number would be represented by a fraction of natural numbers $\frac{p}{q}$. The other number would be irrational, and it would be represented by n .

$$n * \frac{p}{q} = -2 \quad (1.69)$$

$$n * p = -2 * q \quad (1.70)$$

$$n = \frac{-2 * q}{p} \quad (1.71)$$

Since p and q are both natural numbers, that means that $\frac{q}{p}$ is a rational number. Since rationals are closed under multiplication, this means that n is a rational number. This contradicts our initial assumption. This means that $\sqrt{3} + \sqrt{5} \in \mathbb{Q} \iff \sqrt{3} - \sqrt{5} \in \mathbb{Q}$. OEΔ

Now, we prove that neither are rational.

Proof by Contradiction. We will assume for contradiction purposes that both are rational.

$$\sqrt{3} + \sqrt{5} \in \mathbb{Q} \tag{1.72}$$

$$\sqrt{3} - \sqrt{5} \in \mathbb{Q} \tag{1.73}$$

Since rational numbers are closed under addition, we can add the two together and say that the result is rational.

$$(\sqrt{3} - \sqrt{5}) + (\sqrt{3} + \sqrt{5}) = 2\sqrt{3} \rightarrow 2\sqrt{3} \in \mathbb{Q} \tag{1.74}$$

Since 2 is rational and rationals are closed under division (except for 0), then we can say that our result divided by 2 is rational.

$$\frac{2\sqrt{3}}{2} = \sqrt{3} \rightarrow \sqrt{3} \in \mathbb{Q} \tag{1.75}$$

This is a contradiction of an established statement. This means that our initial statement is false. Combining this with the fact that $\sqrt{3} + \sqrt{5} \in \mathbb{Q} \iff \sqrt{3} - \sqrt{5} \in \mathbb{Q}$, we can say that $\sqrt{3} + \sqrt{5} \notin \mathbb{Q}$ and $\sqrt{3} - \sqrt{5} \notin \mathbb{Q}$. OEΔ
QED

1.10 Problem 1.10

Prove that the multiplicative identity in a field is unique.

Solution

We'll prove this by contradiction.

Proof by Contradiction. Suppose that the field has two multiplicative identity elements. The first is 1. Suppose that we call the second one h and that $h \neq 1$. For h to be a multiplicative identity, it would mean that $h \in \mathbb{F}$ and for all $a \in \mathbb{F}$, $a \cdot h = a$. Suppose we were to multiply our two multiplicative identities together. Due to its nature covered in the identity law, $h \cdot 1 = h$. However, our earlier statement would lead us to believe that $h \cdot 1 = 1$. This would only be possible if $h = 1$. This would give us a contradiction, which would mean that the multiplicative identity would be unique. QED $\square$

1.11 Problem 1.11

Given an ordered field $\mathbb{F}$, recall that we defined the positive elements to be a nonempty subset $P \subseteq \mathbb{F}$ that satisfies both of the following conditions:

- (i) If $a, b \in P$, then $a + b \in P$ and $a \cdot b \in P$;
 - (ii) If $a \in \mathbb{F}$ and $a \neq 0$, then either $a \in P$ or $-a \in P$, but not both.
- (a) Give an example of some $P_1 \subseteq \mathbb{R}$ that satisfies (i) but not (ii).
(b) Give an example of some $P_2 \subseteq \mathbb{R}$ that satisfies (ii) but not (i).

1.11.1 Solution (a)

For the set of integers in the range $[-\infty, \infty]$, (i) is satisfied, but not (ii).

1.11.2 Solution (b)

For the set of integers in the range $[1, 2]$, (ii) is satisfied, but not (i).

1.12 Exercise 1.12

Assume that $\mathbb{F}$ is an ordered field and $a, b, c, d \in \mathbb{F}$ with $a < b$ and $c < d$.

(a) Show that $a + c < b + d$.

(b) Prove that it is not necessarily true that $ac < bd$.

Note whenever you use an axiom.

1.12.1 Solution (a)

Proof. We know that if $a < b$ and $c < d$, then $b - a \in P$ and $d - c \in P$. By the order axiom,

$$(b - a) + (d - c) \in P \quad (1.76)$$

$$(b + d) - (a + c) \in P \quad (1.77)$$

By the definition of the $<$ operator,

$$a + c < b + d \quad (1.78)$$

QED

OE Δ

1.12.2 Solution (b)

Counterexample:

$$a = -1; b = 1 \implies a < b \quad (1.79)$$

$$c = -2; d = 2 \implies c < d \quad (1.80)$$

$$a * c = 2; b * d = 2 \implies ac = bd \quad (1.81)$$

This satisfies the case of a counterexample.

1.13 Exercise 1.13

Let a, b and ε be elements of an ordered field.

(a) Show that if $a < b + \varepsilon$ for every $\varepsilon > 0$, then $a \leq b$.

(b) Use part (a) to show that if $|a - b| < \varepsilon$ for all $\varepsilon > 0$, then $a = b$.

Note whenever you use an axiom.

1.13.1 Solution (a)

Proof by Contradiction. Suppose that $b < a$. We also know that $\varepsilon > 0$. These both would themselves mean that $a - b \in P$ and $\varepsilon \in P$. We can put the two of these together by the order axiom, as well as establish some other things.

$$a - b + \varepsilon \in P \quad (1.82)$$

$$a - b > 0 \quad (1.83)$$

It could be that $\varepsilon = \frac{a-b}{2}$. Since $\varepsilon > 0$, this is applicable. Our given information tells us about further relationships.

$$a < b + \varepsilon \quad (1.84)$$

By the order axiom.

$$b + \varepsilon - a \in P \quad (1.85)$$

By the associative law.

$$\varepsilon - (a - b) \in P \quad (1.86)$$

By the order axiom.

$$a - b < \varepsilon \quad (1.87)$$

By an equivalency.

$$a - b < \frac{a - b}{2} \quad (1.88)$$

This is a contradiction of itself.

QED

OEΔ

1.13.2 Solution (b)

Proof. We have a known equation.

$$|a - b| < \varepsilon \quad (1.89)$$

By Note 1.11.g.

$$-\varepsilon < a - b < \varepsilon \quad (1.90)$$

This gets split into two equations.

$$a - b < \varepsilon \quad (1.91)$$

$$-\varepsilon < a - b \quad (1.92)$$

By Note 1.9.a, adding either b or $b + \varepsilon$, respectively.

$$a < b + \varepsilon \quad (1.93)$$

$$b < a + \varepsilon \quad (1.94)$$

We can apply part (a).

$$a \leq b \quad (1.95)$$

$$b \leq a \quad (1.96)$$

There is only case where this does not self-contradict.

$$a = b \quad (1.97)$$

QED

OEΔ

1.14 Exercise 1.14

Prove that the equality $|ab| = |a| \cdot |b|$ holds for all real numbers a and b .

Solution

Proof by Cases. Case 1: $a, b \in P$

$$a = |a| \tag{1.98}$$

$$b = |b| \tag{1.99}$$

$$|ab| = a * b = ab \tag{1.100}$$

$$|a| \cdot |b| = a \cdot b = ab \tag{1.101}$$

As desired.

Case 2: $a \in P, b \notin P$ This applies equally for $a \notin P, b \in P$.

$$|a| = a \tag{1.102}$$

$$|b| = -b \tag{1.103}$$

$$|a| * |b| = a * (-b) = -ab \tag{1.104}$$

Since $a > 0$ and $b < 0$:

$$ab < 0 \tag{1.105}$$

$$|ab| = -ab \tag{1.106}$$

As desired.

Case 3: $a, b \notin P$

$$|a| = -a \tag{1.107}$$

$$|b| = -b \tag{1.108}$$

$$|a||b| = ab \tag{1.109}$$

$$\tag{1.110}$$

Since $a < 0$ and $b < 0$:

$$ab > 0 \tag{1.111}$$

$$|ab| = ab \tag{1.112}$$

As desired.

QED

OEΔ

1.15 Exercise 1.15

For each of the following, find all numbers x which satisfy the expression.

(a) $|x - 4| = 7$

(e) $|x - 1| + |x + 1| > 1$

(b) $|x - 4| < 7$

(f) $|x - 4| > 1$

(c) $|x + 2| < 7$

(g) $|x - 1| \cdot |x + 1| = 0$

(d) $|x - 1| + |x - 2| > 1$

(h) $|x - 1| \cdot |x + 2| = 3$

1.15.1 Solution (a) $x = 4$ and $x = -3$

Case 1: $|x - 4| = x - 4$

$$x - 4 = 7 \quad (1.113)$$

$$x = 11 \quad (1.114)$$

Case 2: $|x - 4| = -(x - 4)$

$$-(x - 4) = 7 \quad (1.115)$$

$$4 - x = 7 \quad (1.116)$$

$$x = -3 \quad (1.117)$$

1.15.2 Solution (b) $x \in (-3, 7)$

Case 1: $|x - 4| = x - 4$

$$x - 4 < 7 \quad (1.118)$$

$$x < 11 \quad x \in (-\infty, 7) \quad (1.119)$$

Case 2: $|x - 4| = -(x - 4)$

$$-(x - 4) < 7 \quad (1.120)$$

$$x - 4 > -7 \quad (1.121)$$

$$x > -3 \quad x \in (-3, \infty) \quad (1.122)$$

Combination:

$$x \in (-\infty, 7) \text{ and } x \in (-3, \infty) \quad (1.123)$$

$$x \in (-\infty, 7) \cap (-3, \infty) \quad (1.124)$$

$$x \in (-3, 7) \quad (1.125)$$

1.15.3 Solution (c) $-9 < x < 5$

There is one point where an absolute value would change at $x = -2$ that divides the field in two.

Case 1: $x < -2$

In this instance, $|x + 2| = -(x + 2) = -x - 2$.

$$|x + 2| < 7 \quad (1.126)$$

$$-x - 2 < 7 \quad (1.127)$$

$$-9 < x \quad (1.128)$$

Case 2: $x > -2$

In this instance, $|x + 2| = x + 2$

$$|x + 2| < 7 \quad (1.129)$$

$$x + 2 < 7 \quad (1.130)$$

$$x < 5 \quad (1.131)$$

Consolidation

Putting our two cases together, we get $\boxed{-9 < x < 5}$.

1.15.4 Solution (d) $x \in (-\infty, -2)$ and $x \in (2, \infty)$

There are two points where the absolute values would change, at $x = 1$ and $x = 2$, which divides the field in three. We can consider all three of these cases.

Case 1: $x < 1$

In this instance, $|x - 1| = -(x - 1) = 1 - x$ and $|x - 2| = -(x - 2) = 2 - x$.

$$|x - 1| + |x - 2| > 1 \quad (1.132)$$

$$1 - x + 2 - x > 1 \quad (1.133)$$

$$3 - 2x > 1 \quad (1.134)$$

$$4 > 2x \quad (1.135)$$

$$x < 2 \quad (1.136)$$

Case 2: $1 < x < 2$

In this instance, $|x - 1| = x - 1$ and $|x - 2| = -(x - 2) = 2 - x$.

$$|x - 1| + |x - 2| > 1 \quad (1.137)$$

$$x - 1 + 2 - x > 1 \quad (1.138)$$

$$1 > 1 \quad (1.139)$$

This is false, so it cannot be the case.

Case 3: $x > 2$

In this instance, $|x - 1| = x - 1$ and $|x - 2| = x - 2$.

$$|x - 1| + |x - 2| > 1 \quad (1.140)$$

$$x - 1 + x - 2 > 1 \quad (1.141)$$

$$2x - 3 > 1 \quad (1.142)$$

$$2x > 4 \quad (1.143)$$

$$x > 2 \quad (1.144)$$

Consolidation

This gives a final range for x .

$$\boxed{x \in (-\infty, -2) \text{ and } x \in (2, \infty)} \quad (1.145)$$

1.16 Exercise 1.16

Let $\max\{x, y\}$ denote the maximum of the real numbers x and y , and let $\min\{x, y\}$ denote the minimum. For example, $\min\{-1, 4\} = -1$, and also $\min\{-1, -1\} = -1$. Prove that

$$\max\{x, y\} = \frac{x + y + |y - x|}{2} \text{ and } \min\{x, y\} = \frac{x + y - |y - x|}{2}$$

Then, find the formula for $\max\{x, y, z\}$ and $\min\{x, y, z\}$.

Solution

Proof by Cases. There are three cases to cover: $x > y$, $x < y$, and $x = y$.

Case 1: $x > y$

This would mean that $|y - x| = -(y - x) = x - y$. Our goal case would be that $\max\{x, y\} = x$ and that $\min\{x, y\} = y$.

$$\max\{x, y\} = \frac{x + y + |y - x|}{2} = \frac{x + y + x - y}{2} = \frac{2x + 0y}{2} = x \quad (1.146)$$

For maximum, this works.

$$\min\{x, y\} = \frac{x + y - |y - x|}{2} = \frac{x + y + y - x}{2} = \frac{0x + 2y}{2} = y \quad (1.147)$$

For minimum, this works. Conclusively, Case 1 works for both minimum and maximum.

Case 2: $x < y$

This would mean that $|y - x| = y - x$. Our goal case would be that $\max\{x, y\} = y$ and that $\min\{x, y\} = x$.

$$\max\{x, y\} = \frac{x + y + |y - x|}{2} = \frac{x + y + y - x}{2} = \frac{0x + 2y}{2} = y \quad (1.148)$$

For maximum, this works.

$$\min\{x, y\} = \frac{x + y - |y - x|}{2} = \frac{x + y - y + x}{2} = \frac{2x + 0y}{2} = x \quad (1.149)$$

For minimum, this works. Conclusively, Case 2 works for both minimum and maximum.

Case 3: $x = y$

This would mean that $|y - x| = 0$. Our goal would be that the maximum and minimum would be either of the results with substitution and neither without.

$$\max\{x, y\} = \frac{x + y + |y - x|}{2} = \frac{x + y}{2} \quad (1.150)$$

$$\max\{x, y\} = \frac{2x}{2} = x \quad (1.151)$$

$$\max\{x, y\} = \frac{2y}{2} = y \quad (1.152)$$

For maximum, this works.

$$\min\{x, y\} = \frac{x + y - |y - x|}{2} = \frac{x + y}{2} \quad (1.153)$$

$$\min\{x, y\} = \frac{2x}{2} = x \quad (1.154)$$

$$\min\{x, y\} = \frac{2y}{2} = y \quad (1.155)$$

For minimum, this works. Conclusively, Case 3 works for both maximum and minimum.

Q.E.D.

OEΔ

For finding the maximum and minimum values of three numbers, we can know that $\max\{x, y, z\} = \max\{\max\{x, y\}, z\}$ and that $\min\{x, y, z\} = \min\{\min\{x, y\}, z\}$.

$$\max\{x, y, z\} = \max\{\max\{x, y\}, z\} \quad (1.156)$$

$$= \frac{\max\{x, y\} + z + |z - \max\{x, y\}|}{2} \quad (1.157)$$

$$= \frac{\frac{x+y+|y-x|}{2} + z + \left| z - \frac{x+y+|y-x|}{2} \right|}{2} \quad (1.158)$$

$$= \frac{x + y + |y - x| + 2z + 2 \left| z - \frac{x+y+|y-x|}{2} \right|}{2} \quad (1.159)$$

$$= \boxed{\frac{x + y + |y - x| + 2z + |2z - x - y - |y - x||}{4}} \quad (1.160)$$

Case 1: $|y - x| = y - x$

$$\max\{x, y, z\} = \frac{x + y + y - x + 2z + |2z - x - y - y + x|}{4} \quad (1.161)$$

$$= \frac{2y + 2z + |2z - 2y|}{4} \quad (1.162)$$

$$= \frac{y + z + |z - y|}{2} = \max\{y, z\} \quad (1.163)$$

Case 2: $|y - x| = -(y - x) = x - y$

$$\max\{x, y, z\} = \frac{x + y + x - y + 2z + |2z - x - y - x + y|}{4} \quad (1.164)$$

$$= \frac{2x + 2z + |2z - 2x|}{4} \quad (1.165)$$

$$= \frac{x + z + |z - x|}{2} = \max\{x, z\} \quad (1.166)$$

$\min\{x, y, z\}$ can be found in a similar way.

1.17 Exercise 1.17

Prove that if $a, b \in R$ and $0 < a < b$, then $a^n < b^n$ for any positive integer n .

Solution

Proof by Induction. Induction comes in two parts, firstly from the smallest value. The smallest value here would be $n = 1$.

$$a^n < b^n \quad (1.167)$$

$$a^1 < b^1 \quad (1.168)$$

$$a < b \quad (1.169)$$

This is a given and confirms the base case requirement.

Next, we have to prove that $a^n < b^n$ implies $a^{n+1} < b^{n+1}$. We can reduce the value on one of the sides to 1.

$$a^n < b^n \quad (1.170)$$

$$a^n(a^n)^{-1} < (a^n)^{-1}b^n \quad (1.171)$$

$$1 < (a^n)^{-1}b^n \quad (1.172)$$

We can do the same with $a < b$, our base case and given.

$$a < b \quad (1.173)$$

$$a \cdot b^{-1} < b \cdot b^{-1} \quad (1.174)$$

$$a \cdot b^{-1} < 1 \quad (1.175)$$

The transitive property can be applied here.

$$a \cdot b^{-1} < (a^n)^{-1}b^n \quad (1.176)$$

$$a^n a \cdot b^{-1} \cdot b < a^n (a^n)^{-1} b^n \cdot b \quad (1.177)$$

$$a^n \cdot a < b^n \cdot b \quad (1.178)$$

$$a^{n+1} < b^{n+1} \quad (1.179)$$

This confirms the successive requirement. As a whole, this confirms the statement.

Q.E.D.

OEΔ

1.18 Exercise 1.18

Prove that if $a_1, a_2, \dots, a_n$ are real numbers, then

$$|a_1 + a_2 + \dots + a_n| \leq |a_1| + |a_2| + \dots + |a_n|$$

Solution

Proof by Induction. Before we begin, we can rewrite our target equation.

$$|a_1 + a_2 + \dots + a_n| \leq |a_1| + |a_2| + \dots + |a_n| \quad (1.180)$$

$$\left| \sum_{i=1}^n a_i \right| \leq \sum_{i=1}^n |a_i| \quad (1.181)$$

Case 1: Minimum value of $n = 1$

We have a value for what would be the left side of the equation.

$$|a_1| \quad (1.182)$$

By the reflexive property, we can create equation for it.

$$|a_1| = |a_1| \quad (1.183)$$

This stands in concurrence with the requirement of the base case.

Case 2: $n \implies n + 1$

We can start with the value of the function for n .

$$|a_1 + a_2 + \dots + a_n| \leq |a_1| + |a_2| + \dots + |a_n| \quad (1.184)$$

$$\left| \sum_{i=1}^n a_i \right| \leq \sum_{i=1}^n |a_i| \quad (1.185)$$

We can add $|a_{n+1}|$ to both sides.

$$\left| \sum_{i=1}^n a_i \right| + |a_{n+1}| \leq \sum_{i=1}^n |a_i| + |a_{n+1}| \quad (1.186)$$

We can take the left side and apply the triangle inequality to it.

$$\left| \sum_{i=1}^n a_i + a_{n+1} \right| \leq \left| \sum_{i=1}^n a_i \right| + |a_{n+1}| \quad (1.187)$$

We can then use the transitive property.

$$\left| \sum_{i=1}^n a_i + a_{n+1} \right| \leq \sum_{i=1}^n |a_i| + |a_{n+1}| \quad (1.188)$$

Put the like terms into sums.

$$\left| \sum_{i=1}^{n+1} a_i \right| \leq \sum_{i=1}^{n+1} |a_i| \quad (1.189)$$

This can be rewritten.

$$|a_1 + a_2 + \cdots + a_n + a_{n+1}| \leq |a_1| + |a_2| + \cdots + |a_n| + |a_{n+1}| \quad (1.190)$$

This is indeed our target to prove by induction. Q.E.D.

OE Δ

1.19 Exercise 1.19

Prove that $\sum_{k=1}^n \frac{1}{k(k+1)} = \frac{n}{n+1}$ for every natural number n .

Solution

Proof by Induction. This is a proof by induction, so we will prove a base case, then prove that every successive number's success implies it works for the next.

Case 1: Base case $n = 1$

$$\sum_{k=1}^n \frac{1}{k(k+1)} = \frac{n}{n+1} \quad (1.191)$$

$$\sum_{k=1}^1 \frac{1}{k(k+1)} = \frac{1}{1+1} \quad (1.192)$$

$$\frac{1}{1(1+1)} = \frac{1}{1+1} \quad (1.193)$$

$$\frac{1}{2} = \frac{1}{2} \quad (1.194)$$

The base case works.

Case 2: $n \implies n+1$

$$\sum_{k=1}^n \frac{1}{k(k+1)} = \frac{n}{n+1} \quad (1.195)$$

We can add the value of $\frac{1}{(n+1)((n+1)+1)}$ to both sides.

$$\sum_{k=1}^n \frac{1}{k(k+1)} + \frac{1}{(n+1)((n+1)+1)} = \frac{n}{n+1} + \frac{1}{(n+1)((n+1)+1)} \quad (1.196)$$

$$\sum_{k=1}^{n+1} \frac{1}{k(k+1)} = \frac{n}{n+1} + \frac{1}{(n+1)(n+2)} \quad (1.197)$$

We can now focus on the term on the right side.

$$\frac{n}{n+1} + \frac{1}{(n+1)(n+2)} \quad (1.198)$$

$$\frac{1}{n+1} \left(n + \frac{1}{n+2} \right) \quad (1.199)$$

$$\frac{1}{n+1} \left(\frac{n(n+2)+1}{n+2} \right) \quad (1.200)$$

$$\frac{n^2+2n+1}{(n+1)(n+2)} \quad (1.201)$$

$$\frac{(n+1)^2}{(n+1)(n+2)} \quad (1.202)$$

$$\frac{n+1}{n+2} \quad (1.203)$$

$$\frac{n+1}{(n+1)+1} \quad (1.204)$$

As such, we can get a final equation.

$$\sum_{k=1}^{n+1} \frac{1}{k(k+1)} = \frac{n+1}{(n+1)+1} \quad (1.205)$$

This is the value we were looking to get, proving the statement by (weak) induction.

Q.E.D.

OEΔ

1.20 Exercise 1.20

Determine which natural numbers, n , have the property that $\sqrt{n}$ is irrational.

Solution

All numbers that are not perfect squares.

1.21 Exercise 1.21

Solution

Given: $f : X \rightarrow Y; A_1, A_2 \subseteq X$ Prove: $f(A_1 \cap A_2) \subseteq f(A_1) \cap f(A_2)$

Proof.

$$f(A_1 \cap A_2) = \{f(x) : x \in A_1 \text{ and } x \in A_2\} \quad (1.206)$$

$$f(A_1) = \{f(x) : x \in A_1\} \quad (1.207)$$

$$f(A_2) = \{f(x) : x \in A_2\} \quad (1.208)$$

Suppose $x \in A_1 \cap A_2$.

$$x \in A_1 \text{ and } x \in A_2 \quad (1.209)$$

$$f(x) \in f(A_1 \cap A_2) \quad (1.210)$$

$$f(x) \in f(A_1) \text{ and } f(x) \in f(A_2) \quad (1.211)$$

$$f(x) \in f(A_1) \cap f(A_2) \quad (1.212)$$

$$\therefore f(x) \in f(A_1 \cap A_2) \rightarrow f(x) \in f(A_1) \cap f(A_2) \quad (1.213)$$

$$\therefore f(A_1 \cap A_2) \subseteq f(A_1) \cap f(A_2) \quad (1.214)$$

Q.E.D.

OEΔ

1.22 Exercise 1.22

Solution

Suppose that A is all positive numbers and B is all non-zero negative numbers. If we have the below function, the case holds.

$$f : \mathbb{Z} \rightarrow \mathbb{N} \tag{1.215}$$

$$n \rightarrow n^2 \tag{1.216}$$

1.23 Exercise 1.23

Solution

Given: $A \subseteq B$; A and B are bounded above

Prove: $\sup(A) \leq \sup(B)$

Proof by Contradiction. Suppose for Contradiction that $\sup(A) > \sup(B)$. We know that $\{x : x \in B \text{ and } x > \sup(B)\} = \emptyset$. This in turn means that $x > \sup(B) \rightarrow x \notin B$. Since $\sup(A) > \sup(B)$, $\sup(A) \notin B$. Since $A \subseteq B$, then $\sup(B)$ is an upper bound of B . Since $\sup(A) > \sup(B)$, that would make $\sup(B)$ the least upper bound of A . This contradicts that $\sup(A)$ is the least upper bound of B . Therefore, our assumption is false and $\sup(A) \leq \sup(B)$. Q.E.D. OE Δ

1.24 Exercise 1.24

1.24.1 Solution (a)

We have to prove that $\sup(A) = M$.

Proof. We know that $x \leq M$ for all $x \in A$. This tells us that M is an upper bound of A .

The other half is that for every other upper bound of A (called a), $M \leq a$.

For all upper bounds a and for all $x \in A$, $x \leq a$.

If there exists an upper bound a such that for one or more $x \in A$ and $x > a'$, a' is not an upper bound of A , so M must be less than or equal to a for any other upper bound of A (called a), or $M \leq a$.

This satisfies the requirements to say that $M = \sup(A)$.

Q.E.D.

OEΔ

1.24.2 Solution (b)

Similar format to solution (a).

1.25 Exercise 1.25

Solution

Proof by Induction. Suppose that A_n indicates a set with n members.

Case 1: Base case $n = 1$

A_1 contains 1 object a . As such, $\max(A) = a$, since a is the only member of A .

Case 2: Inductive step $n \rightarrow n+1$

A_n contains a maximum object m . Suppose we add an object k to A_n to form A_{n+1} . If $m \leq k$, then $\max(A) = m$, so nothing changes and $\max(A) \in A$. If $m > k$, then $\max(A) = k$, so since $k \in A_{n+1}$, $\max(A) \in A$. OE Δ

1.26 Exercise 1.37

(a) Determine

$$\{1, 3, 5\} \cdot \{-3, 0, 1\}$$

(b) Give an example of sets A and B where $\sup(A \cdot B) \neq \sup(A) \cdot \sup(B)$.

Chapter 2

Theories of Cantor (Somehow not Jewish)

2.1 Exercise 2.1

- (a) List all the elements of $\mathcal{P}(\{a, b, c\})$.
- (b) Define a formula for the number of elements in the power set of an n-element set.

2.1.1 Solution (a)

The power set is the set of all subsets. Below is a list of all such subsets.

1	2	3
{a}	{b}	{c}
4	5	6
{a,b}	{a,c}	{b,c}
7	8	
{a,b,c}	$\emptyset$	

There are $\boxed{8}$ members of $\mathcal{P}(\{a, b, c\})$.

2.1.2 Solution (b)

We can look at the first couple values.

n	0	1	2	3
$\mathcal{P}$	1	2	4	8

As it stands, it appears that the proper formula for it is 2^n .

2.2 Exercise 2.2

Prove that $|\{e^n : n \in \mathbb{N}\}| = |\mathbb{N}|$.

Solution

Proof. Let there be a function f .

$$f : \mathbb{N} \rightarrow e^{\mathbb{N}} \tag{2.1}$$

$$n \mapsto e^n \tag{2.2}$$

This pairs every natural number with a number in the left set. Conclusively, the two sets have the same cardinality and there is a bijection. This can be written in math terms.

$$|\{e^n : n \in \mathbb{N}\}| = |\mathbb{N}| \tag{2.3}$$

Q.E.D.

OEΔ

2.3 Exercise 2.3

The following pairs of sets have the same size, and so there exists a bijection between them. Write down an explicit bijection in each case. You do not need to prove your answers.

- | | |
|--------------------------------------|--|
| (a) $(0, \infty)$ and $(1, \infty)$ | (e) $\mathbb{R}$ and $(0, 1)$ |
| (b) $(0, \infty)$ and $(-\infty, 3)$ | (f) $\mathbb{Z}$ and $\{\dots, \frac{1}{8}, \frac{1}{4}, \frac{1}{2}, 1, 2, 4, 8, \dots\}$ |
| (c) $(0, \infty)$ and $(0, 1)$ | (g) $\{0, 1\} \times \mathbb{N}$ and $\mathbb{N}$ |
| (d) $\mathbb{R}$ and $(0, \infty)$ | (h) $[0, 1]$ and $(0, 1)$ |

2.3.1 Solution (a)

$$f : (0, \infty) \rightarrow (1, \infty) \quad (2.4)$$

$$n \mapsto n + 1 \quad (2.5)$$

$$f : (1, \infty) \rightarrow (0, \infty) \quad (2.6)$$

$$n \mapsto n - 1 \quad (2.7)$$

2.3.2 Solution (b)

$$f : (0, \infty) \rightarrow (-\infty, 3) \quad (2.8)$$

$$n \mapsto -n + 3 \quad (2.9)$$

$$f : (-\infty, 3) \rightarrow (0, \infty) \quad (2.10)$$

$$n \mapsto -n + 3 \quad (2.11)$$

2.3.3 Solution (c)

$$f : (0, \infty) \rightarrow (0, 1) \quad (2.12)$$

$$n \mapsto \frac{1}{n+1} \quad (2.13)$$

$$f : (0, 1) \rightarrow (0, \infty) \quad (2.14)$$

$$n \mapsto \frac{1}{n} - 1 \quad (2.15)$$

2.3.4 Solution (d)

$$f : \mathbb{R} \rightarrow (0, \infty) \quad (2.16)$$

$$n \mapsto e^x \quad (2.17)$$

$$f : (0, \infty) \rightarrow \mathbb{R} \quad (2.18)$$

$$n \mapsto \ln(x) \quad (2.19)$$

2.3.5 Solution (e)

$$f : \mathbb{R} \rightarrow (0, 1) \quad (2.20)$$

$$n \mapsto \frac{1}{2n-1} \quad (2.21)$$

$$f : (0, 1) \rightarrow \mathbb{R} \quad (2.22)$$

$$n \mapsto \frac{1}{2n} + \frac{1}{2} \quad (2.23)$$

2.3.6 Solution (f)

$$f : \mathbb{Z} \rightarrow \{..., \frac{1}{8}, \frac{1}{4}, \frac{1}{2}, 1, 2, 4, 8, ...\} \quad (2.24)$$

$$n \mapsto 2^n \quad (2.25)$$

$$f : \{..., \frac{1}{8}, \frac{1}{4}, \frac{1}{2}, 1, 2, 4, 8, ...\} \rightarrow \mathbb{Z} \quad (2.26)$$

$$n \mapsto \log_2(n) \quad (2.27)$$

2.3.7 Solution (g)

$$f : \mathbb{N} \rightarrow \{0, 1\} \times \mathbb{N} \quad (2.30)$$

$$f : \{0, 1\} \times \mathbb{N} \rightarrow \mathbb{N} \quad (2.28)$$

$$(a, b) \mapsto a \cdot b \quad (2.29)$$

$$n \mapsto \left(n \bmod 2, \frac{n - (n \bmod 2)}{2} \right) \quad (2.31)$$

2.3.8 Solution (h)

Iffy solution.

$$f : [0, 1] \rightarrow (0, 1) \quad (2.32)$$

$$n \mapsto \begin{cases} \frac{1}{2} & \text{if } n = 0 \\ \frac{1}{2^{x+2}} & \text{if } n = \frac{1}{2^x} \\ n & \text{otherwise} \end{cases} \quad (2.33)$$

$$f : (0, 1) \rightarrow [0, 1] \quad (2.34)$$

$$n \mapsto \begin{cases} 0 & \text{if } n = \frac{1}{2} \\ \frac{1}{2^{x-2}} & \text{if } n = \frac{1}{2^x} \\ n & \text{otherwise} \end{cases} \quad (2.35)$$

2.4 Exercise 2.4

This problem shows that “equinumerosity is an equivalence relation.” (This justifies the notation $|A| = |B|$.) Let A , B , and C be sets. For this problem only, we’ll write $A \sim B$ to mean that A and B are equinumerous, meaning that there is a bijection $A \rightarrow B$.

- (a) Show that $A \sim A$.
- (b) Show that if $A \sim B$ then $B \sim A$.
- (c) Show that if $A \sim B$ and $B \sim C$, then $A \sim C$.

2.4.1 Solution (a)

We can always establish a bijection between A and A .

$$f : A \rightarrow A \tag{2.36}$$

$$n \mapsto n \tag{2.37}$$

This establishes the bijection.

2.4.2 Solution (b)

It is given that $A \sim B$. We can extract from that that the function from A to B is bijective, and consequently that it is reversible. Said reversability can give us the function from B to A , which would itself be bijective. This bijectivity says that $B \sim A$.

2.4.3 Solution (c)

Since $A \sim B$, then $A \rightarrow B$ is bijective. The same can be done to conclude that $B \rightarrow C$ is bijective. Since both are bijective, this means that $f(a) = f(b) \implies a = b$ and $f(b) = f(c) \implies b = c$, as well as that $b \in B \implies \exists a \in A : f(a) = b$ and $c \in C \implies \exists b \in B : f(b) = c$. By transitivity, since both $A \rightarrow B$ and $B \rightarrow C$ are bijective, then $A \rightarrow C$ is bijective. This in turn means that $A \sim C$.

2.5 Exercise 2.5

- (a) Prove that if A and B are countable sets, then $A \cup B$ is also a countable set.
- (b) Prove that if A_n is a countable set for each $n \in \mathbb{N}$, then the set $\bigcup_{n=1}^{\infty} A_n$ is also countable.

2.5.1 Solution (a)

We can establish a bijective function.

$$f : \mathbb{N} \rightarrow A \cup B \quad (2.38)$$

$$n \rightarrow \begin{cases} A_{\lfloor \frac{x}{2} \rfloor} & \text{if } x \equiv 0 \pmod{2} \\ B_{\lfloor \frac{x}{2} \rfloor} & \text{if } x \equiv 1 \pmod{2} \end{cases} \quad (2.39)$$

$$f : A \cup B \rightarrow \mathbb{N} \quad (2.40)$$

$$(A \cup B)_n \rightarrow n \quad (2.41)$$

This bijection establishes that $A \cup B$ is countable.

2.5.2 Solution (b)

This can be done using the winding bijection, detailed in the textbook. The table for this is below.

	1	2	3	4	...
A_1	a_{11}	a_{21}	a_{31}	a_{41}	...
A_2	a_{12}	a_{22}	a_{32}	a_{42}	...
A_3	a_{13}	a_{23}	a_{33}	a_{43}	...
A_4	a_{14}	a_{24}	a_{34}	a_{44}	...
$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\ddots$

This visual proof establishes the solution.

2.6 Exercise 2.6

Show that $|\mathbb{N}| = |\mathbb{Z}|$ by finding an explicit bijection from $\mathbb{N}$ to $\mathbb{Z}$. You do not need to prove your bijection works.

Solution

$$f : \mathbb{N} \rightarrow \mathbb{Z} \tag{2.42}$$

$$n \rightarrow \lfloor \frac{n}{2} \rfloor (-1)^n \tag{2.43}$$

2.7 Exercise 2.7

Let $A, B \subseteq \mathbb{R}$. We define $A \cdot B = \{a \cdot b : a \in A \text{ and } b \in B\}$.

- (a) Give an example of sets A_1 and B_1 where $|A_1 \cdot B_1| < \max\{|A_1|, |B_1|\}$.
- (b) Give an example of sets A_2 and B_2 where $|A_2 \cdot B_2| > \max\{|A_2|, |B_2|\}$.
- (c) Give an example of sets A_3 and B_3 where $|A_3 \cdot B_3| = \max\{|A_3|, |B_3|\}$.
- (d) For which of the above does there exist an example where one or both of the sets are infinite?

2.7.1 Solution (a)

$$A_1 = \mathbb{N} \tag{2.44}$$

$$B_1 = \{0\} \tag{2.45}$$

$$A_1 \cdot B_1 = \{0\} \tag{2.46}$$

2.7.2 Solution (b)

$$A_2 = \{n : \mathbb{N} \text{ and } 0 < n < 100\} \tag{2.47}$$

$$B_2 = \{-1, 1\} \tag{2.48}$$

$$A_2 \cdot B_2 = \{-99, -98, \dots, -1, 1, 2, \dots, 98, 99\} \tag{2.49}$$

2.7.3 Solution (c)

$$A_3 = \{1, 2, 3, 4, 5\} \tag{2.50}$$

$$B_3 = \{1\} \tag{2.51}$$

$$A_3 \cdot B_3 = \{1, 2, 3, 4, 5\} \tag{2.52}$$

2.7.4 Solution (d)

For (a) and (c), yes.

2.8 Exercise 2.8

- (a) Describe a way to partition the set $\mathbb{N}$ into 6 subsets, each containing infinitely many elements.
(b) Describe a way to partition the set $\mathbb{N}$ into infinitely many subsets, each containing infinitely many elements.

2.8.1 Solution (a)

Divide each number by 6 and take the remainder. The value of each remainder determines the set it would go in. The way to define each set is as follows:

$$f : \mathbb{N} \rightarrow N_1 \qquad n \rightarrow (n - 1) * 6 + 1 \qquad (2.53)$$

$$f : \mathbb{N} \rightarrow N_2 \qquad n \rightarrow (n - 1) * 6 + 2 \qquad (2.54)$$

$$f : \mathbb{N} \rightarrow N_3 \qquad n \rightarrow (n - 1) * 6 + 3 \qquad (2.55)$$

$$f : \mathbb{N} \rightarrow N_4 \qquad n \rightarrow (n - 1) * 6 + 4 \qquad (2.56)$$

$$f : \mathbb{N} \rightarrow N_5 \qquad n \rightarrow (n - 1) * 6 + 5 \qquad (2.57)$$

$$f : \mathbb{N} \rightarrow N_6 \qquad n \rightarrow (n - 1) * 6 + 6 \qquad (2.58)$$

2.8.2 Solution (b)

Create one set S_1 and add one object: 1. Create a second set S_2 . Add 2 to S_1 and 3 to S_2 . Create S_3 . Add 4 to S_1 , 5 to S_2 , and 6 to S_3 . Your sets should look like the below.

$$S_1 = \{1, 2, 4\} \qquad (2.59)$$

$$S_2 = \{3, 5\} \qquad (2.60)$$

$$S_3 = \{6\} \qquad (2.61)$$

A pattern should be discernable here. We can define each individual set.

$$f : \mathbb{N} \rightarrow S_i \qquad (2.62)$$

$$n \rightarrow \sum_{x=0}^i (x) + \sum_{x=i}^{i+n-1} (x) \qquad (2.63)$$

2.9 Exercise 2.9

Is $|\mathbb{Z} \times \mathbb{N}|$ countable or uncountable?

Solution

Previously mentioned is the digonalization and pairing up practice, which has established $\mathbb{Q}$ as being countably infinite. In this instance, we can do the same, but instead of fractions, we can put $\mathbb{N}$ on one side and $\mathbb{Z}$ on the other. In the center will be the pairing up of the numbers. This gives us the below table.

	0	1	-1	2	-2	...
1	{0,1}	{1,1}	{-1,1}	{2,1}	{-2,1}	...
2	{0,2}	{1,2}	{-1,2}	{2,2}	{-2,2}	...
3	{0,3}	{1,3}	{-1,3}	{2,3}	{-2,3}	...
4	{0,4}	{1,4}	{-1,4}	{2,4}	{-2,4}	...
$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\ddots$

We can subsequently use the winding bijection on this to get our countably infinite set. As a note, this can be done in turn with any pair of countably infinite sets paired together in some way, as long as they are mapped in terms of some function $f(a, b)$.

2.10 Exercise 2.10

Let S be the set of sequences (a_n) where, for each $n, a_n \in \{0, 1\}$. Is S countable or uncountable?

Solution

Note: Each sequence is infinite in this case.

Cantor's proof that $\mathbb{R}$ is uncountably infinite may be applicable here. Suppose that we laid out an array of all the sequences as binary decimals less than 1.

$$0.a_{11}a_{12}a_{13}a_{14}a_{15} \quad (2.64)$$

$$0.a_{21}a_{22}a_{23}a_{24}a_{25} \quad (2.65)$$

$$0.a_{31}a_{32}a_{33}a_{34}a_{35} \quad (2.66)$$

$$0.a_{41}a_{42}a_{43}a_{44}a_{45} \quad (2.67)$$

$$0.a_{51}a_{52}a_{53}a_{54}a_{55} \quad (2.68)$$

Suppose that we were to create a member of this set. We could take every value of each member of each set such that the first designating number matches the second designating number (e.g. $a_{11}, a_{22}, a_{33}, a_{xx}, \dots$). For each value of that, we would create a formula for the respective value of member b_i .

$$a_{ii} \rightarrow b_i = \begin{cases} 1 & \text{if } a_{ii} = 0 \\ 0 & \text{if } a_{ii} = 1 \end{cases} \quad (2.69)$$

This in turn creates a new value of a set in S , which is previously unwritten. This in turn establishes that the set is uncountably infinite.

2.11 Exercise 2.11

Suppose that X is a nonempty set. Prove that the following three assertions are equivalent:

- (a) X is finite or countably infinite.
- (b) There is a one-to-one function $f : X \rightarrow \mathbb{N}$.
- (c) There is an onto function $g : \mathbb{N} \rightarrow X$.

Solution

Proof. We can rewrite (b) as for $a, b \in X$ and given $f(a), f(b) \in \mathbb{N}$, the below applies.

$$f(a) = f(b) \implies a = b \quad (2.70)$$

We can rewrite (c) as for $b \in X$, the below applies.

$$\exists a \in \mathbb{N} : g(a) = b \quad (2.71)$$

We can rewrite (a) as there being a correspondence between $\mathbb{N}$ and X such that every . OEΔ

2.12 Exercise 2.12

- (a) Give an example of a collection of countably many disjoint open intervals, or prove that this does not exist.
- (b) Give an example of a collection of uncountably many disjoint open intervals, or prove that this does not exist.

2.13 Exercise 2.21

Prove that the set of points on the unit circle in $\mathbb{R}^2$ (that is, $\{(x, y) : x^2 + y^2 = 1\}$) is uncountable.

Solution

Proof. As given in the definition of a unit circle, there is a formula that (x, y) must follow.

$$x^2 + y^2 = 1 \quad (2.72)$$

This equation can be solved for y .

$$y = \pm\sqrt{1 - x^2} \quad (2.73)$$

This in turn creates a bijection between $\mathbb{R}$ and either $\mathbb{R} \times [0, \infty)$ or $\mathbb{R} \times (-\infty, 0]$, for which we can focus on the former. This bijection can have an associated function.

$$f : \mathbb{R} \rightarrow \mathbb{R} \times [0, \infty) \quad (2.74)$$

$$x \mapsto (x, \sqrt{1 - x^2}) \quad (2.75)$$

For this, there should be a correction that $\sqrt{1 - x^2}$ will only be real if $-1 \leq x \leq 1$, which we can use to correct our function.

$$f : [-1, 1] \rightarrow [-1, 1] \times [0, 1] \quad (2.76)$$

$$x \mapsto (x, \sqrt{1 - x^2}) \quad (2.77)$$

This gives us a domain for the points on a quarter of a unit circle of $[-1, 1]$. It has been previously established that there are uncountable real numbers between 0 and 1. This means that there are uncountable real numbers between -1 and 1. As such, there are unlimited real numbers that are points on a unit circle.

Q.E.D.

OEΔ

2.14 Exercise 2.22

A real number x is said to be *algebraic* (over the rationals) if it satisfies some polynomial equation (of positive degree)

$$a_n x^n + a_{n-1} x^{n-1} + a_{n-2} x^{n-2} + \cdots + a_1 x + a_0 = 0$$

where each $a_i \in \mathbb{Q}$. If a real number is not algebraic, then it is *transcendental*.

(a) Prove that there are countably many algebraic numbers. (You may use the fundamental theorem of algebra which says that a polynomial with degree n has at most n real roots.)

(b) Prove that there are uncountably many transcendental numbers.

2.14.1 Solution (a)

Proof. The fundamental theorem of algebra says that a polynomial of degree n has at most n real roots. This means that the real roots of any given polynomial can be expressed as a finite set of real numbers and as such is countable. This means that any given polynomial has countably many real roots, all of which would be algebraic real numbers.

It has been previously established that that the set of all polynomials with rational coefficients (as we are dealing with presently) is countable. We can in turn count the algebraic numbers by counting each polynomial and counting each of their roots as we count them. This establishes these algebraic numbers as countable.

Q.E.D.

OEΔ

2.14.2 Solution (b)

We can prove this in a similar way to proving there are uncountably many irrational numbers.

Proof by Contradiction. Suppose for contradiction that the set of transcendental numbers T is countable. The set of algebraic numbers A is already established to be countable. We can express this in terms of cardinalities.

$$|A| = |T| = |\mathbb{N}| \tag{2.78}$$

Transcendental numbers are defined as being real numbers that are not algebraic.

$$T = \mathbb{R} \setminus A \quad (2.79)$$

$$A \cup T = \mathbb{R} \quad (2.80)$$

Since both A and T are countable, we can form a function to combine the two through $\mathbb{N}$ which becomes $\mathbb{R}$ given $A \cup T = \mathbb{R}$.

$$f : \mathbb{N} \rightarrow \mathbb{R} \quad (2.81)$$

$$n \mapsto \begin{cases} A(\lceil \frac{n}{2} \rceil) & \text{if } n \equiv 0 \pmod{2} \\ T(\lceil \frac{n}{2} \rceil) & \text{if } n \equiv 1 \pmod{2} \end{cases} \quad (2.82)$$

This establishes that $\mathbb{R}$ is countably infinite. This contradicts that $\mathbb{R}$ is uncountably infinite. As such, we can conclude that there are uncountably many transcendental numbers. OE Δ

Chapter 3

A Game of Hide and Sequence

3.1 Exercise 3.1

Suppose that a sequence (a_n) converges to 0.001. Prove that finitely many values of a_n are negative.

3.1.1 Solution

Proof. Recall the definition of convergence.

$$\forall \varepsilon > 0, \exists N : \forall n > N, |a_n - 0.001| < \varepsilon \quad (3.1)$$

The last part of that can have the absolute value removed.

$$-\varepsilon < a_n - 0.001 < \varepsilon \quad (3.2)$$

Suppose that $\varepsilon = 0.001$.

$$-0.001 < a_n - 0.001 < 0.001 \quad (3.3)$$

$$0 < a_n < 0.002 \quad (3.4)$$

This means that for all $n > N$, a_n is positive. This means that a finite number N of a_n can be negative or zero. This means that a finite number of a_n are negative. OEΔ

3.2 Exercise 3.22

Suppose (a_n) and (b_n) are sequences.

- (a) Prove that $a_n \rightarrow L$ and $a_n \leq M$ for all n , then $L \leq M$.
- (b) Assume that $a_n \leq b_n$ for all n . Prove that if $a_n \rightarrow L$ and $b_n \rightarrow M$, then $L \leq M$.

3.2.1 Solution (b)

Proof by Contradiction. Suppose for contradiction that $L > M$.

We know from the start that $a_n \rightarrow L$ and $b_n \rightarrow M$.

$$\forall \varepsilon > 0, \exists N : \forall n > N, |a_n - L| < \varepsilon \quad (3.5)$$

$$\forall \varepsilon > 0, \exists N : \forall n > N, |b_n - M| < \varepsilon \quad (3.6)$$

Furthermore, we can subtract b_n from a_n , which also has a limit.

$$b_n - a_n \rightarrow M - L \quad (3.7)$$

$$\forall \varepsilon > 0, \exists N : \forall n > N, |(b_n - a_n) - (M - L)| < \varepsilon \quad (3.8)$$

In both (3.5) and (3.6), $\varepsilon > 0$. We can recall $L > M$, just so it's enumerated.

$$L > M \quad (3.9)$$

$$L - M > 0 \quad (3.10)$$

$$|L - M| = L - M \quad (3.11)$$

Take (3.6) and manipulate it into a similar equation by multiplying by -1 in the absolute value (it changes nothing else).

$$|M - b_n| < \varepsilon \quad (3.12)$$

We can add together (3.5) and (3.12) and get another $\varepsilon > 0$.

$$|a_n - L| + |M - b_n| < \varepsilon \quad (3.13)$$

We can apply the triangle inequality here.

$$|a_n - b_n - L + M| \leq |a_n - L| + |M - b_n| < \varepsilon \quad (3.14)$$

Next, apply the inverse triangle inequality.

$$||a_n - b_n| - |L - M|| \leq |a_n - b_n - L + M| < \varepsilon \quad (3.15)$$

Recall the statement that $a_n \leq b_n$ and manipulate that.

$$a_n \leq b_n \quad (3.16)$$

$$0 \leq b_n - a_n \quad (3.17)$$

$$|b_n - a_n| = |a_n - b_n| = b_n - a_n \quad (3.18)$$

Now, we can substitute in values for (3.15) elated to absolute values, specifically recalling (3.11) and (3.18).

$$|(b_n - a_n) - (L - M)| < \varepsilon \quad (3.19)$$

This in turn implies that $b_n - a_n \rightarrow L - M$, which is a contradiction of (3.7). Ergo, $L \leq M$. OE Δ

3.32 Exercise 3.32

Let $(a_n) \subset \mathbb{R}$. Prove that if every subsequence of (a_n) converges, then (a_n) does.

3.32.1 Solution

Proof. Suppose that we create a subsequence of (a_n) that includes every member of (a_n) except the first. By the rule, it converges.

$$\forall \varepsilon > 0, \exists N \in \mathbb{N} \setminus \{1\} : \forall n > N, |a_n - a| < \varepsilon \quad (3.20)$$

Suppose that we do add a_1 to the sequence as its first member. There would be no violation of the rules of convergence. This would also be the sequence (a_n) exactly. Therefore, (a_n) converges. OEΔ

3.33 Exercise 3.33

3.34 Exercise 3.34

Give an example of two sequences (a_n) and (b_n) that satisfy the following four conditions.

- (a) $\lim_{n \rightarrow \infty} a_n$ does not exist.
- (b) $\lim_{n \rightarrow \infty} b_n$ does not exist.
- (c) $(a_n \cdot b_n) \rightarrow \infty$
- (d) $(a_n \cdot b_{n+1}) \rightarrow -\infty$

3.34.1 Solution

Suppose we define (a_n) and (b_n) .

$$a_n := (-1)^n n \tag{3.21}$$

$$b_n := (-1)^{n+1} n \tag{3.22}$$

3.35 Exercise 3.36

Define a sequence (a_n) recursively as follows. Let a_1 and a_2 be a pair of real numbers, and define recursively for all $n \geq 3$.

$$a_n := \frac{a_{n-1} + a_{n-2}}{2}$$

Does (a_n) necessarily converge?

3.35.1 Solution

Begin with (a_n) .

$$a_n := \frac{a_{n-1} + a_{n-2}}{2} \quad (3.23)$$

By necessity, this is the average of the two and less than or equal to one of the prior values but greater than or equal to the other.

Compute $|\delta a_n| = |a_n - a_{n-1}|$.

$$|\delta a_n| = |a_n - a_{n-1}| = \left| \frac{a_{n-1} + a_{n-2}}{2} - a_{n-1} \right| = \left| \frac{a_{n-2} - a_{n-1}}{2} \right| \quad (3.24)$$

Let's check if $|\delta a_n|$ is monotone decreasing. Start with finding the value of δa_{n+1} .

$$\delta a_{n+1} = \frac{a_{n-1} - a_n}{2} = \frac{a_{n-1} - \frac{a_{n-1} + a_{n-2}}{2}}{2} \quad (3.25)$$

$$= \frac{2a_{n-1} - a_{n-1} - a_{n-2}}{4} = \frac{a_{n-1} - a_{n-2}}{4} \quad (3.26)$$

This proof will involve some math, just to prove that $(|\delta a_n|)$ converges to 0, which will be done right now. It largely starts with a simple inequality of halves and absolute values.

$$\left| \frac{a_{n-1} - a_{n-2}}{2} \right| \leq |a_{n-1} - a_{n-2}| \quad (3.27)$$

$$\left| \frac{a_{n-1} - a_{n-2}}{4} \right| \leq \left| \frac{a_{n-1} - a_{n-2}}{2} \right| \quad (3.28)$$

$$|\delta a_{n+1}| \leq |\delta a_n| \quad (3.29)$$

($|\delta a_n|$) is monotone decreasing. We also can, from what we did here, develop an equation for $|\delta a_n|$ for $n \geq 3$.

$$|\delta a_n| = \left(\frac{1}{2}\right)^{n-2} |\delta a_2| \quad (3.30)$$

Since $|\delta a_2|$ is a constant and $\left(\frac{1}{2}\right)^{n-2} \rightarrow 0$, then $|\delta a_n| \rightarrow 0$.

$$\forall \varepsilon > 0, \exists N : \forall n > N, |\delta a_n| < \varepsilon \quad (3.31)$$

Substituting in the definition of $|\delta a_n|$ gives us something further.

$$\forall \varepsilon > 0, \exists N : \forall n > N, |a_n - a_{n-1}| < \varepsilon \quad (3.32)$$

We can use the triangle inequality to have this work for any $m, n > N$.

$$|a_n - a_{n-1}| < \varepsilon \quad (3.33)$$

$$|a_{n+1} - a_n| < \varepsilon \quad (3.34)$$

$$|a_{n+1} - a_{n-1}| \leq |a_{n+1} - a_n| + |a_n - a_{n-1}| < \varepsilon \quad (3.35)$$

This can be expanded to work for any $m, n > N$.

$$\forall \varepsilon > 0, \exists N : \forall m, n > N, |a_m - a_n| < \varepsilon \quad (3.36)$$

This in turn means that (a_n) is Cauchy. This means that (a_n) necessarily converges.

3.36 Problem 3.38

Give an example of a monotone sequence that is not Cauchy.

3.36.1 Solution

Define (a_n) .

$$a_n := n \quad (3.37)$$

It is monotonically increasing, but it diverges.

3.37 Problem 3.40

For each of the following, give an example of a sequence with that property or prove no such sequences exist.

- (a) A Cauchy sequence with an unbounded subsequence.
- (b) An unbounded sequence with a Cauchy subsequence.

3.37.1 Solution (a)

Proof. Recall the definition of a Cauchy sequence.

$$\forall \varepsilon > 0, \exists N : \forall m, n > N, |a_m - a_n| < \varepsilon \quad (3.38)$$

We can rewrite the last part of this.

$$-\varepsilon < a_m - a_n < \varepsilon \quad (3.39)$$

$$a_n - \varepsilon < a_m < a_n + \varepsilon \quad (3.40)$$

This means that a_m is bounded by the other possible a_n . There are values not bounded by this restriction, the ones where $m \leq N$. Suppose we found the maximum value of a_m where $m \leq N$.

$$S = \{a_1, a_2, \dots, a_N\} \quad (3.41)$$

Suppose we add $a_n + \varepsilon$ and $a_n - \varepsilon$ to this and set $\varepsilon = 1$.

$$\begin{aligned} S &= \{a_1, a_2, \dots, a_N, a_n - \varepsilon, a_n + \varepsilon\} \\ &= \{a_1, a_2, \dots, a_N, a_n - 1, a_n + 1\} \end{aligned} \quad (3.42)$$

We can define a maximum U and a minimum L value of this, which set bounds for (a_n) .

$$U = \max(S) \quad (3.43)$$

$$L = \min(S) \quad (3.44)$$

As such, for all m , $L \leq a_m \leq U$. Suppose we call our subsequence (a_{n_k}) . Members of (a_{n_k}) will all be members of (a_n) . This means that for all n_k , $L \leq a_{n_k} \leq U$. This means that it is bounded. OEΔ

3.37.2 Solution (b)

I will define a sequence whose Cauchy subsequence is only the odd terms.

$$a_n := n^{(-1)^n} \tag{3.45}$$

3.38 Problem 3.42

Suppose the sequence (x_n) is the sum of two other sequences, (y_n) and (z_n) . That is $x_n = y_n + z_n$ for all n . If (x_n) is bounded, and (y_n) and (z_n) are both monotone, must (x_n) be convergent? What if (y_n) and (z_n) are also bounded?

3.38.1 Solution (a): Unbounded

In the first case, (x_n) must not necessarily be convergent. Suppose we define two new sequences.

$$\delta y_n := y_{n+1} - y_n \quad (3.46)$$

$$\delta z_n := z_{n+1} - z_n \quad (3.47)$$

From here, we can define δx_n .

$$\delta x_n = \delta y_n + \delta z_n = y_{n+1} - y_n + z_{n+1} - z_n = x_{n+1} - x_n \quad (3.48)$$

For (x_n) to converge, (x_n) must be Cauchy.

$$\forall \varepsilon > 0, \exists N : \forall m, n > N, |x_m - x_n| < \varepsilon \quad (3.49)$$

It's possible that $m = n + 1$.

$$|x_{n+1} - x_n| = |\delta x| = |\delta y_n + \delta z_n| < \varepsilon \quad (3.50)$$

Now, suppose that $\delta y_n = \delta z_n = \frac{\varepsilon}{2}$.

$$|\delta x| = |\delta y_n + \delta z_n| = \left| \frac{\varepsilon}{2} + \frac{\varepsilon}{2} \right| = |\varepsilon| = \varepsilon \quad (3.51)$$

This would not be possible if (x_n) has to be convergent. Therefore, (x_n) does not have to be convergent.

3.38.2 Solution (b): Bounded

Recall the monotone convergence theorem. It says that if a sequence is monotone, it converges if and only if it is bounded. This means that (y_n) and (z_n) are also Cauchy.

$$\forall \varepsilon > 0, \exists N_y : \forall m, n > N_y, |y_m - y_n| < \varepsilon \quad (3.52)$$

$$\forall \varepsilon > 0, \exists N_z : \forall m, n > N_z, |z_m - z_n| < \varepsilon \quad (3.53)$$

We can add the last terms of these two together, then use the Triangle Inequality. In this case, set $m, n > N = \max\{N_y, N_z\}$.

$$\forall \varepsilon > 0, \exists N_y : \forall m, n > N \quad (3.54)$$

$$\varepsilon > |y_m - y_n| + |z_m - z_n| \geq |y_m - y_n + z_m - z_n| \quad (3.55)$$

$$\varepsilon > |y_m + z_m - y_n - z_n| = |x_m - x_n| \quad (3.56)$$

Put this together with (3.52) or (3.53).

$$\forall \varepsilon > 0, \exists N_y : \forall m, n > N, |x_m - x_n| < \varepsilon \quad (3.57)$$

This means that (x_n) is Cauchy, which means that it converges.

Scratch Work (b)

MCT says y_n and z_n both converge.

$$\forall \varepsilon > 0, \exists N : \forall n > N, |y_n - y| < \varepsilon$$

$$\forall \varepsilon > 0, \exists N : \forall n > N, |z_n - z| < \varepsilon$$

$$|y_n - y| + |z_n - z| < 2\varepsilon$$

$$\text{Cauchy: } 2\varepsilon > |y_m - y_n| + |z_m - z_n| \geq |y_m - y_n + z_m - z_n|$$

$$|y_m - y_n + z_m - z_n| = |y_m + z_m - y_n - z_n| = |x_m - x_n|$$

(x_n) is Cauchy, so it converges.

3.39 Problem 3.44

Prove that if (a_n) is a bounded sequence which does not converge, then it must contain two subsequences, both of which converge, but which converge to different values.

3.39.1 Solution

Proof. Bolzano-Weierstrass says every bounded sequence has a convergent subsequence. Naturally, (a_n) also has one, which we could call (a_{n_k}) . Suppose we call the set of all n_k , $\mathbb{N}_k$. We can create three sets from these two sequences.

$$A = \{a_n | n \in \mathbb{N}\} \quad (3.58)$$

$$A_k = \{a_{n_k} | n_k \in \mathbb{N}_k\} \quad (3.59)$$

$$A_k^c = \{a_n | n \in \mathbb{N} \setminus \mathbb{N}_k\} \quad (3.60)$$

Let $(m_j)_{j=1}^\infty$ be the increasing enumeration of $\mathbb{N} \setminus \mathbb{N}_k$. Then $(a_{m_j})_{j=1}^\infty$ is the sequence of terms of (a_n) not contained in (a_{n_k}) .

For (a_n) not to converge, (a_{m_j}) must be an infinite sequence.

Proof by Contradiction. Suppose for contradiction that (a_{m_j}) is finite. That means that (m_j) is finite. Since it is finite, it has a maximum member, which we call M . For all $m > M$, $a_m \notin (a_{m_j})$. By definition, for all $m > M$, $a_m \in (a_{n_k})$. (a_{n_k}) converges, which we can use the definition of.

$$\forall \varepsilon > 0, \exists N : \forall n_k > N, |a_{n_k} - a| < \varepsilon \quad (3.61)$$

For some values of N , $N < M$, so if $n > M$, then $n > N$. For others, $N > M$, so if $n > N$, then $n > M$. In either case, since (a_{m_j}) converges, we can rewrite its convergence definition to include M .

$$\forall \varepsilon > 0, \exists N : \forall n > \begin{cases} N & \text{if } N > M \\ M & \text{if } M \geq N \end{cases}, |a_n - a| < \varepsilon \quad (3.62)$$

If we add back the values of (a_{m_j}) , this does not change. Quite the opposite, it would say that (a_n) converges. This contradicts that (a_n) does not converge. Ergo, our initial assumption is wrong, and (a_{m_j}) is not finite.

OEΔ

Past that, since (a_n) is bounded, (a_{m_j}) is also bounded. This means that (a_{m_j}) either is convergent or has a convergent subsequence. If (a_{m_j}) converges to the same value as (a_{n_k}) , all subsequences of (a_n) converge to the same number, which would mean that (a_n) converges to that number, which means that (a_n) converges, which we know cannot be the case. That would mean that (a_{m_j}) converges to something else.

If (a_{m_j}) does not converge, it's possible that its convergent subsequence converges to the same value as (a_{n_k}) , but then we'd end up with another bounded, infinite subsequence. That could go on infinitely. If all of these converge to the same number, that means that all subsequences of (a_n) converge to the same number, so (a_n) converges, which is not the case. Ergo, one (or more) of them has to converge to a different number, which gives us two subsequences that converge to different limits. OEΔ

3.40 Problem 3.45

Let (a_n) be the sequence where $a_1 = 1$ and, for each $n > 1$, $a_n = a_{n-1} + \frac{1}{n^2}$. In 1734, Leonhard Euler famously proved that (a_n) converges to $\frac{\pi^2}{6}$. Now let (b_n) be the sequence where $b_1 = 1$ and, for each $n > 1$, $b_n = b_{n-1} + \frac{1}{n^3}$. Use Euler's result to prove that (b_n) converges.

3.40.1 Solution

Proof. Let's redefine a_n and b_n .

$$a_n = a_{n-1} + \frac{1}{n^2} = 1 + \sum_{i=1}^n \frac{1}{i^2} = \sum_{i=1}^n \frac{1}{i^2} \quad (3.63)$$

$$b_n = b_{n-1} + \frac{1}{n^3} = 1 + \sum_{i=1}^n \frac{1}{i^3} = \sum_{i=1}^n \frac{1}{i^3} \quad (3.64)$$

Let's start with just the origin point of that.

$$i \geq 1 \quad (3.65)$$

Let's get closer to our definitions to a_n and b_n .

$$i^3 \geq i^2 \quad (3.66)$$

$$\frac{1}{i^2} \geq \frac{1}{i^3} \quad (3.67)$$

With every increment in n , a_n increases by more than b_n . This means that for all $n > 1$, $a_n > b_n$. This gives (b_n) an upper bound of (a_n) , so (a_n) is bounded.

Furthermore, for all $i > 0$ (and by extension $n > 0$), $\frac{1}{i^3} > 0$, so (b_n) is monotonically increasing. A monotonically increasing sequence bounded above converges. Ergo, (b_n) converges. OEΔ

Scratch Work

$$a_n = 1 + \sum_2^n \frac{1}{i^2}$$

$$b_n = 1 + \sum_2^n \frac{1}{i^3}$$

$$\sum \frac{1}{i^2} \geq \sum \frac{1}{i^3}$$

$$\frac{1}{i^2} \geq \frac{1}{i^3} i^3 \geq i^2$$

$$i \geq 1$$

Chapter 4

Series not on TV

4.1 Problem 4.1

Determine whether each of the following converges conditionally, converges absolutely, or diverges. You do not need to prove your answers, but state which of the following tests gives the answers: the k^{th} -term test, the geometric series test, and the alternating series test.

- | | | |
|---|---|--|
| (a) $\sum_{k=1}^{\infty} (-1)^k \frac{1}{\sqrt{k}}$ | (d) $\sum_{k=1}^{\infty} \frac{1}{k!}$ | (g) $\sum_{k=1}^{\infty} (-1)^k \frac{1}{\sqrt[3]{k}}$ |
| (b) $\sum_{k=1}^{\infty} (-1)^k \frac{k}{k+7}$ | (e) $\sum_{k=1}^{\infty} (\sqrt{k+1} - \sqrt{k})$ | (h) $\sum_{k=53}^{\infty} \frac{\ln(k)}{\ln(\ln(k))}$ |
| (c) $\sum_{k=1}^{\infty} \frac{1}{(\ln(4))^k}$ | (f) $\sum_{k=1}^{\infty} \frac{1}{2}$ | (i) $\sum_{k=7}^{\infty} \left(\frac{2}{k} + \frac{3}{k^2} \right)$ |

4.2 Problem 4.3

Prove that if $\sum_{k=1}^{\infty} |a_k|$ converges, then $\sum_{k=1}^{\infty} a_k$ converge too. This is Proposition 4.18.

4.3 Problem 4.5

Give an example of a series where $\sum_{k=1}^{\infty} a_k$ converges but $\sum_{k=1}^{\infty} a_{2k}$ diverges.

4.4 Problem 4.7

Give an example of a series $\sum_{k=1}^{\infty} a_k$ where:

- $\sum_{k=1}^{\infty} a_k$ converges,
- $\sum_{k=1}^{\infty} a_k^2$ diverges
- $\sum_{k=1}^{\infty} a_k^3$ converges

4.5 Problem 4.9

The geometric series test (Proposition 4.9) says that a geometric series diverges if $|r| \geq 1$. Recall that a series can diverge to ∞ , to $-\infty$, or can be “does not exist.” Which form of divergence is it for a geometric series? Note that your answer may depend on r .

4.6 Problem 4.11

- (a) Find a way to write $77.77777777\dots$ as a geometric series, and then prove this number is rational by using the geometric series test to write this number as a fraction with integers in the numerator and denominator.
- (b) Write $77.77777777\dots$ as a different geometric series, and use the geometric series test to write this number as a fraction with integers in the numerator and denominator. Are your two fractions the same?
- (c) A number q has a repeating decimal if the non-integer portion of its decimal expansion is repetitive. For example, $72.578578578578578\dots$ has a repeating decimal. Prove that if a number q has a repeating decimal, then q is rational.

4.7 Problem 4.13

- (a) Prove that if (ka_k) converges to a nonzero real number L , then the series $\sum_{k=1}^{\infty} a_k$ diverges. Give an example to show that the converse is false.
- (b) Prove that if (k^2a_k) converges (to any real number), then the series $\sum_{k=1}^{\infty} a_k$ converges. Give an example to show that the converse is false.

4.7.1 Solution (a)

Proof. We know that (ka_k) converges to a nonzero L .

$$\forall \varepsilon > 0, \exists N : \forall k > N, |ka_k - L| < \varepsilon \quad (4.1)$$

It is very much possible that $\varepsilon = \frac{|L|}{2}$. We can also apply the reverse triangle inequality and isolate $|a_k|$.

$$\frac{|L|}{2} = \varepsilon > |ka_k - L| \geq ||ka_k| - |L|| \quad (4.2)$$

$$\frac{|L|}{2} > |ka_k| - |L| > -\frac{|L|}{2} \quad (4.3)$$

$$\frac{3|L|}{2} > |ka_k| > \frac{|L|}{2} \quad (4.4)$$

$$\frac{3|L|}{2} > k|a_k| > \frac{|L|}{2} \quad (4.5)$$

$$\frac{3|L|}{2k} > |a_k| > \frac{|L|}{2k} = \frac{|L|}{2} \cdot \frac{1}{k} \quad (4.6)$$

Because of its convergence to L , beyond a certain point, a_k has the same sign as L . If we only work beyond that point, we can remove the absolute values. This gives us two cases. First is the case where $L > 0$.

$$\frac{3L}{2k} > a_k > \frac{L}{2} \cdot \frac{1}{k} \quad (4.7)$$

The other case is the one where $L < 0$.

$$-\frac{3L}{2k} > -a_k > -\frac{L}{2} \cdot \frac{1}{k} \quad (4.8)$$

We can substitute in a new sequence (b_k) in case 2 where for any k , $b_k = -a_k$ to get back to the point described in case 1. Since it's the same point, I won't be replacing a_k with b_k .

$$\frac{3L}{2k} > a_k > \frac{L}{2} \cdot \frac{1}{k} \quad (4.9)$$

Now, recall the harmonic series.

$$\sum_{k=1}^{\infty} \frac{1}{k} \rightarrow \infty \quad (4.10)$$

We can multiply this in both cases by $\frac{L}{2}$.

$$\sum_{k=1}^{\infty} \frac{L}{2} \cdot \frac{1}{k} \rightarrow \infty \quad (4.11)$$

Since the harmonic series diverges and (a_k) is always greater than our above alteration of the harmonic series, $\sum_{k=1}^{\infty} a_k$ diverges. OEΔ

For an example of how the converse could be false, consider the series $\sum_{k=1}^{\infty} k$. It diverges, and the components of it, even when multiplied by k , also diverge.

4.7.2 Solution (b)

Proof. We know that $(k^2 a_k)$ converges to a real number L .

$$\forall \varepsilon > 0, \exists N : \forall k > N, |k^2 a_k - L| < \varepsilon \quad (4.12)$$

We consider this based on whether L is or is not equal to zero.

Case 1: $L \neq 0$. If $L \neq 0$, it is very much possible that $\varepsilon = \frac{|L|}{2}$. We can

also apply the reverse triangle inequality and isolate $|a_k|$.

$$\frac{|L|}{2} = \varepsilon > |k^2 a_k - L| \geq ||k^2 a_k| - |L|| \quad (4.13)$$

$$\frac{|L|}{2} > |k^2 a_k| - |L| > -\frac{|L|}{2} \quad (4.14)$$

$$\frac{3|L|}{2} > |k^2 a_k| > \frac{|L|}{2} \quad (4.15)$$

$$\frac{3|L|}{2} > k^2 |a_k| > \frac{|L|}{2} \quad (4.16)$$

$$\frac{3|L|}{2k^2} = \frac{3|L|}{2} \cdot \frac{1}{k^2} > |a_k| > \frac{|L|}{2k^2} = \frac{|L|}{2} \cdot \frac{1}{k^2} \quad (4.17)$$

By the series p-test, we can recognize the convergence of $\sum_{k=1}^{\infty} \frac{1}{k^2}$, and multiply that by a constant.

$$\sum_{k=1}^{\infty} \frac{1}{k^2} \rightarrow 0 \quad (4.18)$$

$$\sum_{k=1}^{\infty} \frac{3|L|}{2} \cdot \frac{1}{k^2} \rightarrow 0 \quad (4.19)$$

By the comparison test, this means that $\sum_{k=1}^{\infty} |a_k|$ converges. This in turn means that $\sum_{k=1}^{\infty} a_k$ converges.

Now for the other case.

Case 2: $L = 0$. We can rewrite the initial definition of convergence based on this.

$$\forall \varepsilon > 0, \exists N : \forall k > N, |k^2 a_k| < \varepsilon \quad (4.20)$$

Now, suppose that $\varepsilon = 1$. Solve for $|a_k|$.

$$|k^2 a_k| < 1 \quad (4.21)$$

$$|a_k| < \frac{1}{k^2} \quad (4.22)$$

As with earlier, this means the sum converges. OEΔ

For the example disproving the converse, consider $(\sum_{k=1}^{\infty} k^{-\frac{3}{2}})$. This converges. However, $(k^2 a_k)$ is equal to $(\sqrt{k})$, which diverges to infinity.

4.8 Problem 4.15

Prove the Cauchy condensation test. That is, suppose that (a_n) is a decreasing sequence for which $a_k \rightarrow 0$. Prove that $\sum_{k=1}^{\infty} a_k$ converges if and only if $\sum_{k=1}^{\infty} 2^k a_{2^k}$ converges.

4.8.1 Solution

First, prove that if $\sum_{k=1}^{\infty} a_k$ converges, $\sum_{k=1}^{\infty} 2^k a_{2^k}$ converges.

Proof. Start with convergent $\sum_{k=1}^{\infty} a_k$ for decreasing $a_k \rightarrow 0$. We can recall the definition of the two.

$$\forall \varepsilon > 0, \exists N : \forall k > N, |a_k| < \varepsilon \quad (4.23)$$

$$a_{k+1} < a_k \quad (4.24)$$

$$\sum_{k=1}^{\infty} a_k \rightarrow L_1 \quad (4.25)$$

Here, we can try grouping terms in (a_k) . There is an inequality that is extrapolated from this being a decreasing sequence.

$$\forall n \in \mathbb{N}, 2^{n+1} - 1 \geq j \geq 2^n \implies a_{2^n} \geq a_j \geq a_{2^{n+1}-1} \quad (4.26)$$

$$\sum_{k=2^n}^{2^{n+1}-1} a_k \geq 2^n a_{2^{n+1}} \geq 0 \quad (4.27)$$

We can furthermore sum this over n .

$$\sum_{n=1}^{\infty} \sum_{k=2^n}^{2^{n+1}-1} a_k \geq \sum_{n=1}^{\infty} 2^n a_{2^{n+1}} \geq 0 \quad (4.28)$$

The former term is equal to the converging $\sum_{k=1}^{\infty} a_k$. Therefore, the latter term also converges. The latter term can be turned into $\sum_{k=1}^{\infty} 2^k a_{2^k}$ by adding a few terms.

$$\sum_{k=1}^{\infty} 2^k a_{2^k} = 2 * \sum_{n=1}^{\infty} 2^n a_{2^{n+1}} + 2 * a_2 = \sum_{n=1}^{\infty} 2^{n+1} a_{2^{n+1}} + 2 * a_2 \quad (4.29)$$

None of these operations (multiplying by 2 and adding $2 * a_2$) compromise convergence. Ergo, $\sum_{k=1}^{\infty} 2^k a_{2^k}$ converges. OEΔ

Next, prove that if $\sum_{k=1}^{\infty} 2^k a_{2^k}$ converges, then $\sum_{k=1}^{\infty} a_k$ converges.

Proof. We begin with the earlier statement.

$$\forall n \in \mathbb{N}, 2^{n+1} - 1 \geq j \geq 2^n \implies a_{2^n} \geq a_j \geq a_{2^{n+1}-1} \quad (4.30)$$

$$2^n a_{2^n} \geq \sum_{k=2^n}^{2^{n+1}-1} a_k \geq 0 \quad (4.31)$$

Sum this over n .

$$\sum_{n=1}^{\infty} 2^n a_{2^n} \geq \sum_{n=1}^{\infty} \sum_{k=2^n}^{2^{n+1}-1} a_k \geq 0 \quad (4.32)$$

The latter term converges. As described earlier, it is equal to the sum over $\mathbb{N}$ of (a_k) .

$$\sum_{n=1}^{\infty} \sum_{k=2^n}^{2^{n+1}-1} a_k = \sum_{k=1}^{\infty} a_k \quad (4.33)$$

Therefore, $\sum_{k=1}^{\infty} a_k$ converges. OEΔ

4.9 Problem 4.17

Prove the ratio test via the following steps. Given a series $\sum_{k=1}^{\infty} a_k$ with $a_k \neq 0$, assume that

$$\lim_{k \rightarrow \infty} \left| \frac{a_{k+1}}{a_k} \right| =: r < 1$$

We will prove that the series converges absolutely.

- (a) Let q be such that $r < q < 1$. Explain why there is some N such that $n \geq N$ implies that $|a_{n+1}| \leq |a_n| \cdot q$.
- (b) Explain why $\sum_{k=1}^{\infty} |a_N| \cdot q^k$ necessarily converges.
- (c) Finally, use part (b) to prove that $\sum_{k=1}^{\infty} |a_k|$ converges.

4.10 Problem 4.18

Prove the root test via the following steps. Given a series $\sum_{k=1}^{\infty} a_k$ where each $a_k \geq 0$, assume that the limit $\lim_{k \rightarrow \infty} (a_k)^{1/k}$ exists. Call this limit ρ . Then this series converges if $\rho < 1$ and diverges if $\rho > 1$. (The test is inconclusive if $\rho = 1$.)

- (a) Suppose $\rho < 1$. Let $\varepsilon = \frac{1-\rho}{2}$ and $\rho_1 = \rho + \varepsilon$. Prove that there is some N for which $(a_n)^{1/n} < \rho_1$ for all $n > N$.
- (b) Prove that $\sum_{k=N}^{\infty} a_k$ converges by comparing it to a geometric series. Then conclude that $\sum_{k=1}^{\infty} a_k$ also converges.
- (c) Suppose $\rho > 1$. Let $\varepsilon = \frac{\rho-1}{2}$ and $\rho_2 = \rho - \varepsilon$. Prove that there is some N for which $(a_n)^{1/n} > \rho_2$ for all $n > N$.
- (d) Prove that $\sum_{k=N}^{\infty} a_k$ diverges by comparing it to a geometric series. Then conclude that $\sum_{k=1}^{\infty} a_k$ also diverges.

4.10.1 Solution (a)

Begin with the limit from earlier and recall its convergence definition (traditional and Cauchy).

$$\lim_{k \rightarrow \infty} (a_k)^{1/k} = \rho < 1 \quad (4.34)$$

$$\forall \varepsilon > 0, \exists N : \forall n > N, |(a_n)^{1/n} - \rho| < \varepsilon \quad (4.35)$$

$$\forall \varepsilon > 0, \exists N : \forall m, n > N, |(a_m)^{1/m} - (a_n)^{1/n}| < \varepsilon \quad (4.36)$$

Expand the first absolute value, then isolate $(a_n)^{1/n}$.

$$-\varepsilon < (a_n)^{1/n} - \rho < \varepsilon \quad (4.37)$$

$$\rho - \varepsilon < (a_n)^{1/n} < \rho + \varepsilon \quad (4.38)$$

4.11 Problem 4.20

- (a) Give an example of two divergent series $\sum_{k=1}^{\infty} a_k$ and $\sum_{k=1}^{\infty} b_k$, such that $\sum_{k=1}^{\infty} a_k b_k$ converges.
- (b) Give an example of two convergent series $\sum_{k=1}^{\infty} a_k$ and $\sum_{k=1}^{\infty} b_k$, such that $\sum_{k=1}^{\infty} a_k b_k$ diverges.
- (c) Prove that if $\sum_{k=1}^{\infty} a_k$ and $\sum_{k=1}^{\infty} b_k$ converge absolutely, then $\sum_{k=1}^{\infty} a_k b_k$ converges absolutely.

4.11.1 Solution (a)

$$a_k = b_k = \frac{1}{k} \quad (4.39)$$

$$a_k b_k = \frac{1}{k^2} \quad (4.40)$$

4.11.2 Solution (b)

$$a_k = b_k = (-1)^{k+1} \frac{1}{\sqrt{k}} \quad (4.41)$$

$$a_k b_k = \frac{1}{k} \quad (4.42)$$

4.11.3 Solution (c)

Proof. Note that these two converge absolutely. This means that their absolute value sums also converge.

$$\sum_{k=1}^{\infty} |a_k| \rightarrow L_1 \quad (4.43)$$

$$\sum_{k=1}^{\infty} |b_k| \rightarrow L_2 \quad (4.44)$$

Since they converge, we can use the contrapositive of the k^{th} -term test to determine what they converge to.

$$a_k \rightarrow 0; |a_k| \rightarrow 0 \quad (4.45)$$

$$b_k \rightarrow 0; |b_k| \rightarrow 0 \quad (4.46)$$

Take the member absolute value functions and multiply them together.

$$|a_k| |b_k| = |a_k b_k| \rightarrow 0 \quad (4.47)$$

Since they converge to 0, there is a point after which $|a_k| < 1$ and $|b_k| < 1$.

$$\exists N \in \mathbb{N} : \forall k > N, |a_k| < 1 \quad (4.48)$$

$$\exists N \in \mathbb{N} : \forall k > N, |b_k| < 1 \quad (4.49)$$

We can multiply the last term of (4.48) by $|b_k|$ on both sides.

$$|b_k| > |a_k| |b_k| = |a_k b_k| \quad (4.50)$$

By the comparison test, this means that the sum of $(|a_k b_k|)$ over k converges. From that, we can conclude that the sum of $(a_k b_k)$ over k converges.

$$\sum_{k=1}^{\infty} |a_k b_k| \rightarrow L_3 \quad (4.51)$$

$$\sum_{k=1}^{\infty} a_k b_k \rightarrow L_4 \quad (4.52)$$

OEΔ

4.12 Problem 4.22

Consider the sum $\sum_{k=1}^{\infty} a_k$, and define s_n to be this series' n^{th} partial sum; that is, $s_n = \sum_{k=1}^n a_k$. The series $\sum_{k=1}^{\infty} a_k$ is called Cesaro summable if the following converges:

$$\lim_{n \rightarrow \infty} \frac{s_1 + s_2 + \cdots + s_n}{n}$$

- (a) Prove that if $\sum_{k=1}^{\infty} a_k$ converges, then this series is Cesaro summable.
- (b) Prove by example that if $\sum_{k=1}^{\infty} a_k$ is Cesaro summable this does not imply that $\sum_{k=1}^{\infty} a_k$ converges.

4.12.1 Solution (a)

Proof. Since $\sum_{k=1}^{\infty} a_k$ converges, its series of partial sums (a_n) also converges. Recall Problem 3.29 in chapter 3, where we proved that if $(a_n) \rightarrow a$, $b_n := \frac{a_1 + a_2 + \cdots + a_n}{n} \rightarrow a$. Suppose we apply that to (s_n) .

$$s_n \rightarrow s \tag{4.53}$$

$$\frac{s_1 + s_2 + \cdots + s_n}{n} \rightarrow s \tag{4.54}$$

Conclsively, this means that (s_n) converges.

OE Δ

4.12.2 Solution (b)

Take the case of $a_n = (-1)^n$. The sum diverges to a non-existant limit. However, dividing by the number of terms makes it converge to 0.5.

4.13 Problem 4.23

Show that if $\sum_{k=1}^{\infty} a_k$ is conditionally convergent, then there exists a rearrangement of this sum which diverges to ∞ .

4.14 Problem 4.24

Show that if $\sum_{k=1}^{\infty} a_k$ is conditionally convergent, then there exists a rearrangement of this sum whose limit does not exist.

4.15 Problem 4.25

Assume that $\sum_{k=1}^{\infty} a_k$ is conditionally convergent. Define the limit superior of a sequence (s_n) (Notation: “ $\limsup_{n \rightarrow \infty} s_n$ ”) to be

$$\limsup_{n \rightarrow \infty} s_n = \lim_{n \rightarrow \infty} \left(\sup_{m \geq n} s_m \right)$$

And define the limit inferior of a sequence (s_n) (Notation: “ $\liminf_{n \rightarrow \infty} s_n$ ”) to be

$$\liminf_{n \rightarrow \infty} s_n = \lim_{n \rightarrow \infty} \left(\inf_{m \geq n} s_m \right)$$

Prove that for any $\alpha, \beta \in \mathbb{R} \cup \{\pm\infty\}$ with $\alpha \leq \beta$, there is a rearrangement of $\sum_{k=1}^{\infty} a_k$ whose sequence of partial sums, (s_n) , have

$$\limsup_{n \rightarrow \infty} s_n = \beta \quad \text{and} \quad \liminf_{n \rightarrow \infty} s_n = \alpha$$

4.15.1 Solution

Proof. We know that, for a conditionally convergent series, there exists a rearrangement that converges to any $L \in \mathbb{R} \cup \pm\infty$. Suppose defined a new variable δ such that $\delta = \frac{|\alpha - \beta|}{4}$. We can separate a_n into a sequence of its non-negative and negative terms, called (p_n) and (n_n) respectively.

$$\sum_{k=1}^{\infty} p_k \rightarrow \infty \tag{4.55}$$

$$\sum_{k=1}^{\infty} n_k \rightarrow -\infty \tag{4.56}$$

We can order (p_k) and (n_k) such that both converge to 0.

$$\forall \varepsilon > 0, \exists N : \forall k > N, |p_k| < \varepsilon \tag{4.57}$$

$$\forall \varepsilon > 0, \exists N : \forall k > N, |n_k| < \varepsilon \tag{4.58}$$

There exists some P_0 and N_0 that we can sum up to the points P_0 and N_0 that $|p_{P_0}| < |\alpha - \beta|$ and $|n_{N_0}| < |\alpha - \beta|$.

$$\sum_{k=1}^{P_0} p_k + \sum_{k=1}^{N_0} n_k = c_0 \tag{4.59}$$

Beyond that point, there exists some P_1 that we can sum to that will be greater than $\beta - \delta$.

$$\exists P_1 \geq 0 : c_0 + \sum_{k=P_0+1}^{P_1} p_k = b_1 \geq \beta - \delta \quad (4.60)$$

From there, we can sum over (n_n) until the total is less than $\alpha + \delta$.

$$\exists N_1 \geq 0 : \alpha < b_1 + \sum_{k=N_0+1}^{N_1} n_k = c_1 \leq \alpha + \delta \quad (4.61)$$

We can continue this for any $m \geq 1$.

$$\forall m \in \mathbb{N}, \exists P_m \geq P_{m-1} : \beta > c_{m-1} + \sum_{k=P_{m-1}+1}^{P_m} p_k = b_m \geq b_{m-1} \quad (4.62)$$

$$\forall m \in \mathbb{N}, \exists N_m \geq N_{m-1} : c_{m-1} > b_m + \sum_{k=N_{m-1}+1}^{N_m} n_k = c_m \leq \alpha \quad (4.63)$$

Summing this up over all m , we end up with a sum that covers all of (a_n) and which has a limit inferior of α and limit superior of β . OE Δ

4.16 Problem 4.26

Prove that if $\sum_{k=1}^{\infty} a_k$ converges absolutely to L , then any rearrangement of this sum also converges to L .

4.16.1 Solution

4.17 Problem 4.27

Prove that if each $a_k \geq 0$ and $\sum_{k=1}^{\infty} a_k = \infty$, then any rearrangement of this sum also diverges to ∞ .

4.18 Problem 4.29

There is a neat theorem which, in a small way, relies on the fact that the harmonic series $\sum_{k=1}^{\infty} \frac{1}{k}$ diverges. Here's the theorem: Every positive rational number can be written as a finite sum of distinct numbers of the form $1/n$. For example, the positive rational number $\frac{7243}{4140}$ can be written like this:

$$\frac{7243}{4140} = \frac{1}{1} + \frac{1}{2} + \frac{1}{5} + \frac{1}{21} + \frac{1}{527} + \frac{1}{3,054,492}$$

Now, take a moment to appreciate how magical this seems, because once you see the algorithm which produces the decomposition, it might lose some of its shine. Here's how I found the above decomposition of $\frac{7243}{4140}$ (which I basically chose at random):

$$\begin{aligned} \frac{7243}{4140} - 1 &= \frac{3103}{4140} \\ \frac{3103}{4140} - \frac{1}{2} &= \frac{1033}{4140} \\ \frac{1033}{4140} &< \frac{1}{3} \text{ and } \frac{1}{4} \\ \frac{1033}{4140} - \frac{1}{5} &= \frac{41}{828} \\ \frac{41}{828} &< \frac{1}{6}, \frac{1}{7}, \dots, \frac{1}{20} \\ \frac{41}{828} - \frac{1}{21} &= \frac{11}{5796} \\ \frac{11}{5796} &< \frac{1}{22}, \frac{1}{23}, \dots, \frac{1}{526} \\ \frac{11}{5796} - \frac{1}{527} &= \frac{1}{3,054,492} \end{aligned}$$

- (a) If the fraction that we start with is much bigger than 1, like $\frac{244,406,536}{7}$, why is this not an issue? Why will subtracting off 1, and then $1/2$, and then $1/3$, and so on, eventually catch up to it?
- (b) Prove that if this operation is performed on a fraction $\frac{p}{q}$ which is less than 1, then the numerator necessarily decreases (in the above example, the numerators decreased at every stage: 7243, 3103, 1033, 41, 11, 1). Note: The algorithm always subtracts off the biggest possible reciprocal. That is, if $\frac{p}{q} < 1$, then we subtract off $\frac{1}{n}$ where $\frac{1}{n} < \frac{p}{q} < \frac{1}{n-1}$.

(c) Explain why this proves the theorem.

4.18.1 Solution (a)

By subtracting reciprocals of natural numbers, we will be subtracting consecutive terms of the Harmonic Series. The Harmonic series diverges, so for the rational number $\frac{p}{q}$ there will be a point P at which $\sum_{k=1}^P \frac{1}{k} < \frac{p}{q} < \sum_{k=1}^{P+1} \frac{1}{k}$.

4.18.2 Solution (b)

Proof. On any given reciprocal subtracted, there is going to be an initial rational number $\frac{p_1}{q_1}$ which we subtract $\frac{1}{n}$ from to get the unreduced rational number $\frac{p_2}{q_2}$.

$$\frac{p_2}{q_2} = \frac{p_1}{q_1} - \frac{1}{n} = \frac{np_1 - q_1}{nq_1} \quad (4.64)$$

Since $\frac{p_2}{q_2}$ is unreduced, we can say that $q_2 = nq_1$.

$$\frac{p_2}{q_2} = \frac{np_1 - q_1}{nq_1} \quad (4.65)$$

$$p_2 = np_1 - q_1 \quad (4.66)$$

Furthermore, $\frac{p_1}{q_1}$ lies in between two reciprocals of natural numbers, which we can extrapolate on.

$$\frac{1}{n-1} > \frac{p_1}{q_1} \quad (4.67)$$

$$q_1 > np_1 - p_1 \quad (4.68)$$

$$p_1 > np_1 - q_1 \quad (4.69)$$

We can put that into our above equation for p_2 .

$$p_2 < p_1 \quad (4.70)$$

Even if the rational number is reduced, this will still be the case, because the new p_2 will be less than the initial p_2 . OEΔ

4.18.3 Solution (c)

This is because eventually, the rational number will be reduced to a point where $\frac{1}{n} < \frac{p}{q} < \frac{1}{n-1}$. After that point, the numerator will start reducing, and keep reducing until it hits 1.